

Week 10: Capstone Results and Discussion

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1 Capstone Results & Discussion

1.1 Results: IBM Dataset

Here the IBM dataset is pruned to conform to the MARS paradigm, given that the data was heavily skewed towards retention (i.e the target variable had many more instances of retention than attrition), a Python dictionary of class weights is created to force the model's hand in prioritizing attrition. While non-informative columns are dropped as they were with the IBM dataset in this study, the native **profession** column from the Babushkin dataset (which is a multi factor categorical column indicating the specialization of an employee) will be transformed into the binary **hr_profession** variable as exempting this variable from transformation will have compelled the models in this pipeline to split on highly sparse, fragmented categories for the remaining 50% of the company not engaged in HR roles thereby diluting the predictive power of the feature considerably. Hence this transformation forces the analytical pipeline to confront a highly specific binary question: Does being an HR insider structurally alter an employee's retainability by the organizations that employ them?. The forthcoming sections will serve to answer this query.

IBM Dataset Pipeline | Shape Filtered: (1470, 35) -> (1470, 16)
Computed Imbalance Weights: {0: 0.6, 1: 3.1}

Final MARS-Tagged Features:

- JobSatisfaction (M)
- TrainingTimesLastYear (M)
- StockOptionLevel (M)
- PerformanceRating (A)
- Education (A)
- YearsAtCompany (A)
- JobLevel (R)
- YearsWithCurrManager (R)
- YearsInCurrentRole (R)
- DistanceFromHome (S)
- EnvironmentSatisfaction (S)
- EducationField_Human Resources (A)
- EducationField_Life Sciences (A)
- EducationField_Marketing (A)
- EducationField_Medical (A)
- EducationField_Other (A)
- EducationField_Technical Degree (A)
- Department_Human Resources (R)
- Department_Research & Development (R)
- Department_Sales (R)
- OverTime_0 (S)
- OverTime_1 (S)

- BusinessTravel_0 (S)
- BusinessTravel_1 (S)
- BusinessTravel_2 (S)

1.1.1 Confusion Matrix Display & Analysis (IBM Dataset)

Figure 1 below represents a **confusion matrix** which is a visualization that employs a square $N \times N$ matrix summarizing a **Classification model**'s performance by mapping true class labels against their predicted counterparts obtained from the **classifiers**. It also allows for calculating key **performance metrics** that were discussed in the Methods section of this study like **precision**, **recall**, and F_1 **scores** to identify class-specific performance issues.

As such, Figure 1 will serve as an illustrative example of how these key **metrics** are obtained with this technique while future computations of these **scores** will be tabulated in summary tables to enhance interpretability and maintain control over visual overload. In terms of the analytical utility of Figure 1, it can be observed that the results gleaned in this visual serve as a teaching example of the “**Accuracy Trap**” pervasive in **Machine learning** and *attrition* studies. The **Std. Logistic Regression model** here misclassified 28 out of 36 *attriting* employees as being *retainable* while providing a reasonably **accuracy rate** (~85%) that could confound the naive interpreter into overstating the utility of this **model** to drive *retention* in enterprise HR settings.

This failure at once justifies the use of **regularization techniques** in **linear models** and the inclusion of **distance based** and **non-linear models** in our overall methodology to offset these observed eccentricities.

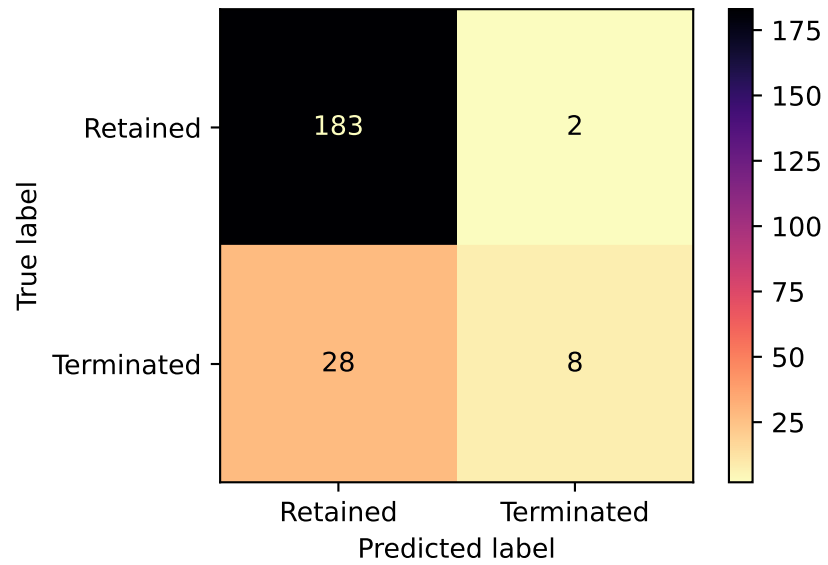


Figure 1: Confusion Matrix for Std Logistic Regression Model (IBM)

The classification report for the A: Standard Logistic Regression model is:

	precision	recall	f1-score	support
0	0.87	0.99	0.92	185
1	0.80	0.22	0.35	36
accuracy			0.86	221
macro avg	0.83	0.61	0.64	221
weighted avg	0.86	0.86	0.83	221

1.1.2 kNN Elbow Diagram & Feature Selection (IBM Dataset)

kNN Classifiers are notoriously prone to having their geometric distance calculations diluted by an overabundance of **model** features, therefore sklearn's **SelectKBest** will be used to shrink our featureset down to the 10 most influential variables, this is accomplished by performing a one-way **Analysis of Variance (ANOVA) F-test** (by using the **f_classif** setting of this function) between each individual numerical feature and the categorical **target variable** to determine their linear dependency and then narrowing down the feature set by selecting Features with the highest resulting **F-statistic scores**.

kNN will now use only these 10 features:

```
['JobSatisfaction (M)' 'StockOptionLevel (M)' 'YearsAtCompany (A)'  
'JobLevel (R)' 'YearsWithCurrManager (R)' 'YearsInCurrentRole (R)'  
'EnvironmentSatisfaction (S)' 'OverTime_0 (S)' 'OverTime_1 (S)'  
'BusinessTravel_2 (S)']
```

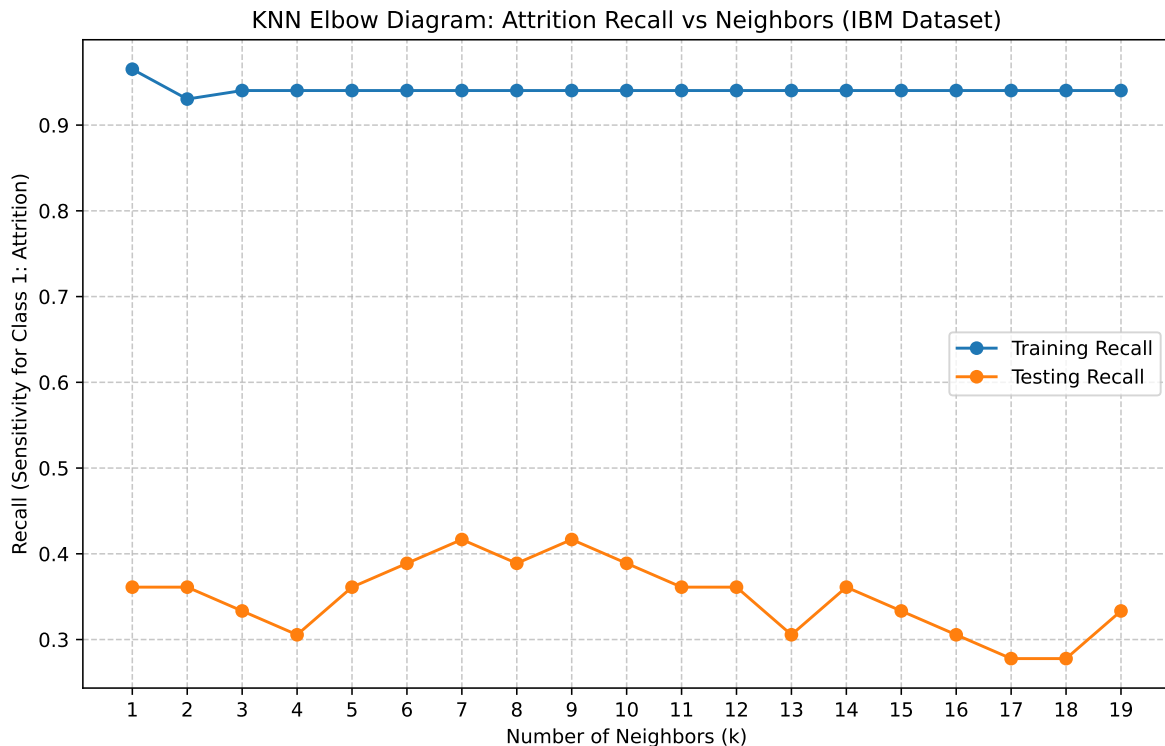


Figure 2: KNN Elbow Diagram: Attrition Recall vs. Neighbors (IBM Dataset)

Peak Performance: k=7 with Test Recall of 0.4167

Observing Figure 2 above, we discover that the **kNN algorithm (Model E)** identified **k = 7** as the optimal number of neighbors, which in the realm of enterprise HR translates to an employee's *flight risk* being best obtained from a “pod” or circle of their 7 closest peers.

1.1.3 Global Feature Importances (VIMS Facet Grid - IBM Dataset)

While the **kNN model** above computes its insights via the construction of peer groups, it is imperative for us to understand the Global implications of the features in overall dataset in affecting *turnover outcomes*, this is accomplished via the following facet grid of feature importances Figure 3

Readily apparent from Figure 3 above, are the elevation of **JobLevel (R)** and **YearsWithCurrManager (R)** features as universal predictors of *retention* (indicated by their representation as blue bars) via our panel of **predictive models**. Conversely **OverTime_1 (S)** (or being called into *Overtime shifts/labor*) is universally flagged as a top *Attrition Driver* (Red bars) or a top magnitude splitter (Grey bars).

This prioritization underscores the very limited utility of *financial or compensatory recourse* to alleviate *attrition* emanating from poorly defined roles (low values of **JobLevel (R)**) or poor or experimental managerial allotment and configuration (demonstrated by low values of **YearsWithCurrManager (R)**)

Also visible here is the impact of **regularization techniques** that reveal themselves in the disparity between the feature importances produced by **Models A & B**. In **Model A** **YearsAtCompany (A)** is a moderate red bar and the #9 overall factor in predicting *attrition*, however in **Model B** this feature ascends in importance to the #1 predictor of *attrition* ostensibly due to the preponderance of multicollinear or correlated features that are endemic to HR datasets (e.g *tenure, time in role* etc). In penalizing this noise with an **L2 penalty**, **Model B** uncovers a harsh HR reality: Working at an organization for extended periods of time without progressing upwards in terms of *Job Level* or *seniority* or being transferred to a different manager leads to *stagnation* and *flight-risk*. In short: Decoupling *tenure* from *job mobility* can have deleterious consequences for *retention*.

The **decision tree (Model D)** offers a unique understanding of *turnover trends* in IBM in its promotion of **OverTime_0** and **JobLevel** as predictive variables, this promotion is computational rather than accidental.*Decision trees **eschew constructing gradual slopes or probabilities in their endeavor to mine insights but instead use** metrics of impurity (gini or log-loss) **to find the cleanest cuts in bifurcating the workforce into stayers and leavers. This is most operational when HR managers require hard-and-fast operational rules to flag flight risks (rather than gradual probabilities) and in this vein** Model D** avers that overtime status (or the lack thereof) provides the cleanest immediate filter.

Non-linear models in the Figure 3 panel, such as the **Logistic GAM (Model F)**, underscore the primacy of targeted retention efforts like stock grants (denoted by **StockOptionLevel (M)**)

Global Feature Importance by Algorithm (IBM Dataset)
 Red = Attrition Driver | Blue = Retention Factor | Grey = Magnitude Only

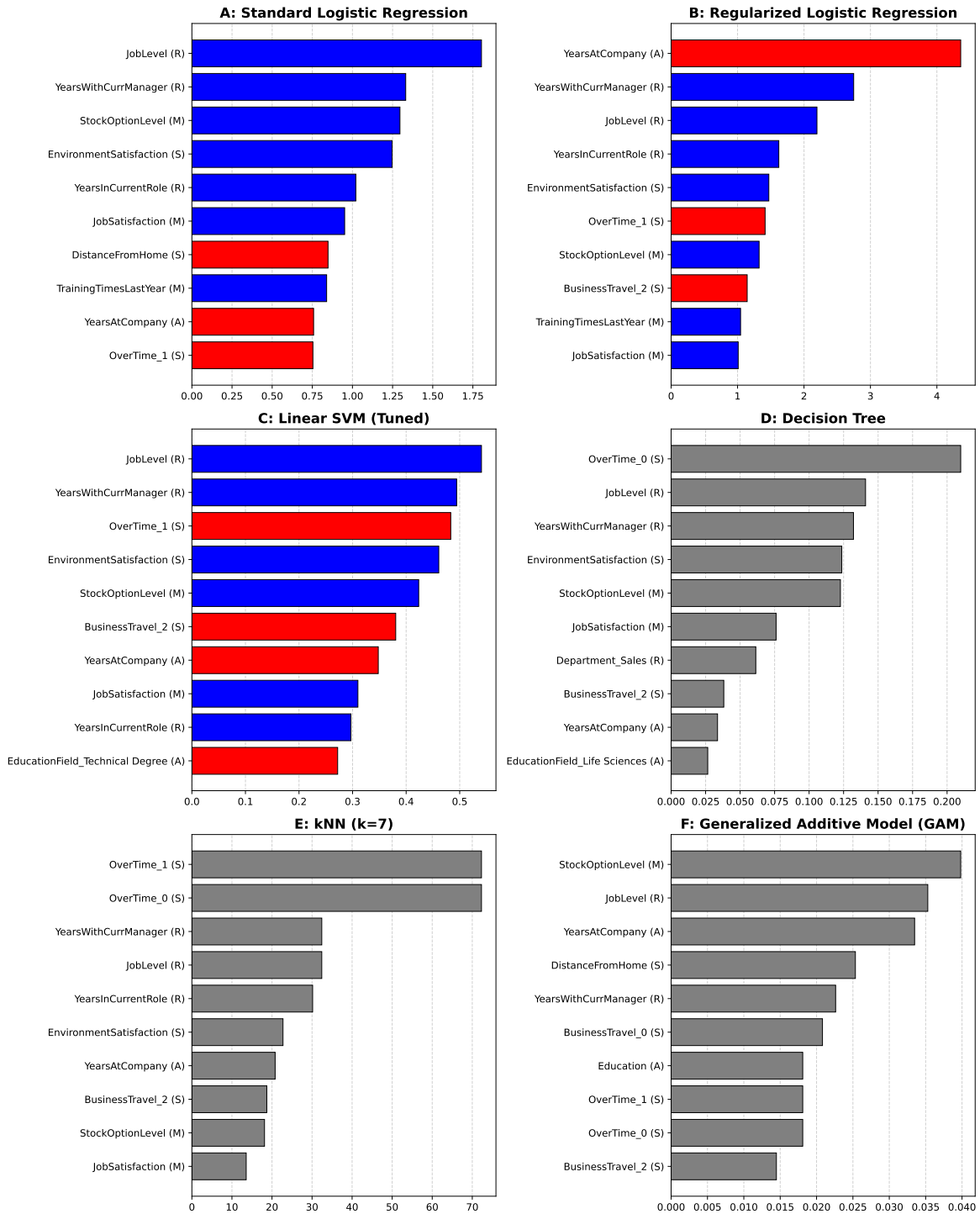


Figure 3: Global Feature Importance by Algorithm (IBM Dataset)

in the IBM dataset, despite only being marginally influential in the linear models of the panel (**Models A through C**). This is likely due to the nature of linear models in forcing straight lines through the dataset; doing so in the context of **StockOptionLevel (M)** implies that every extra stock option confers upon the enterprise the *exact same amount of loyalty*. GAMs dispel this illusion and allow for a nuanced understanding with their usage of splines. The resulting implication here is that extrinsic motivators like compensatory rewards are not flat and can experience *diminishing returns* beyond a certain threshold, which will be explored further in the next section.

1.1.4 Logistic GAM Partial Dependence Plots (IBM Dataset)

he **Partial Dependence Plots (PDPs)** rendered below isolate a single variable to show its exact, marginal effect on an employee's **probability** of quitting, holding all other variables constant. Within each **PDP** diagram, the X-axis represents the employee's status for that specific feature, scaled from 0.0 (lowest possible value in the dataset) to 1.0 (highest possible value). The Y axis represents the **log-odds** of our binary **target variable** (*Attrition* for the datasets employed in this study), most crucial here is the horizontal dashed line at 0.0 which can be understood in lay terms as the “*turnover line*”, therefore any point on the curve above the zero line means that specific value drives the employee to depart the enterprise (higher *attrition risk*) whereas Any point below the zero-line means that value drives them to stay by acting as a *retention anchor*.

Accordingly, the **PDP** plots above in Figure 4 offer us validation in the previously concluded limited utility of *compensatory rewards*, the **StockOptionLevel (M)** plot demonstrates here that conferring a mid-tier *stock grant* on an employee buys the issuing enterprise massive *retention power* but issuing the absolute maximum *stock grant* value squanders enterprise capital in its failure to secure additional *loyalty* in proportion to the elevated stock issuance.

The **JobLevel PDP** plot adds to our nonlinear grasp of IBM data by highlighting the prevalence of a “*Middle-management squeeze*” at IBM, *Attrition risk* starts out high for entry level employees (**JobLevel** = 0.0) drops below the zero line before again resurfacing at a middle range (**JobLevel** between 0.4 and 0.5) before its terminal plunge into *retentive safety* for Senior executives (**JobLevel** >= 0.8). This insight escaped the scrutiny of the **linear models** because they'd averaged out the very same **splines** that produce these insights. **Logistic GAM** reveals here that middle managers at IBM are a considerable *flight risk* ostensibly due to them coming out on the losing end of the *stress to compensation ratio* (i.e they have comparable *workload* to senior leadership sans the lucrative *stock rewards* or the enhanced *autonomy* prevalent at the upper echelons of an enterprise)

The **YearsAtCompany PDP** plot offers its own cautionary tale in the form of demonstrable *tenure fatigue*. It can be observed in Figure 4 that the curve stays below the zero line for people who are in the early or intermediate stages of their *career trajectory* (**YearsAtCompany** between 0.0 and 0.7), but once this threshold is crossed, the **log-odds** favoring *attrition* rocket upward

Non-Linear Feature Effects: Top 3 IBM Drivers (GAM)

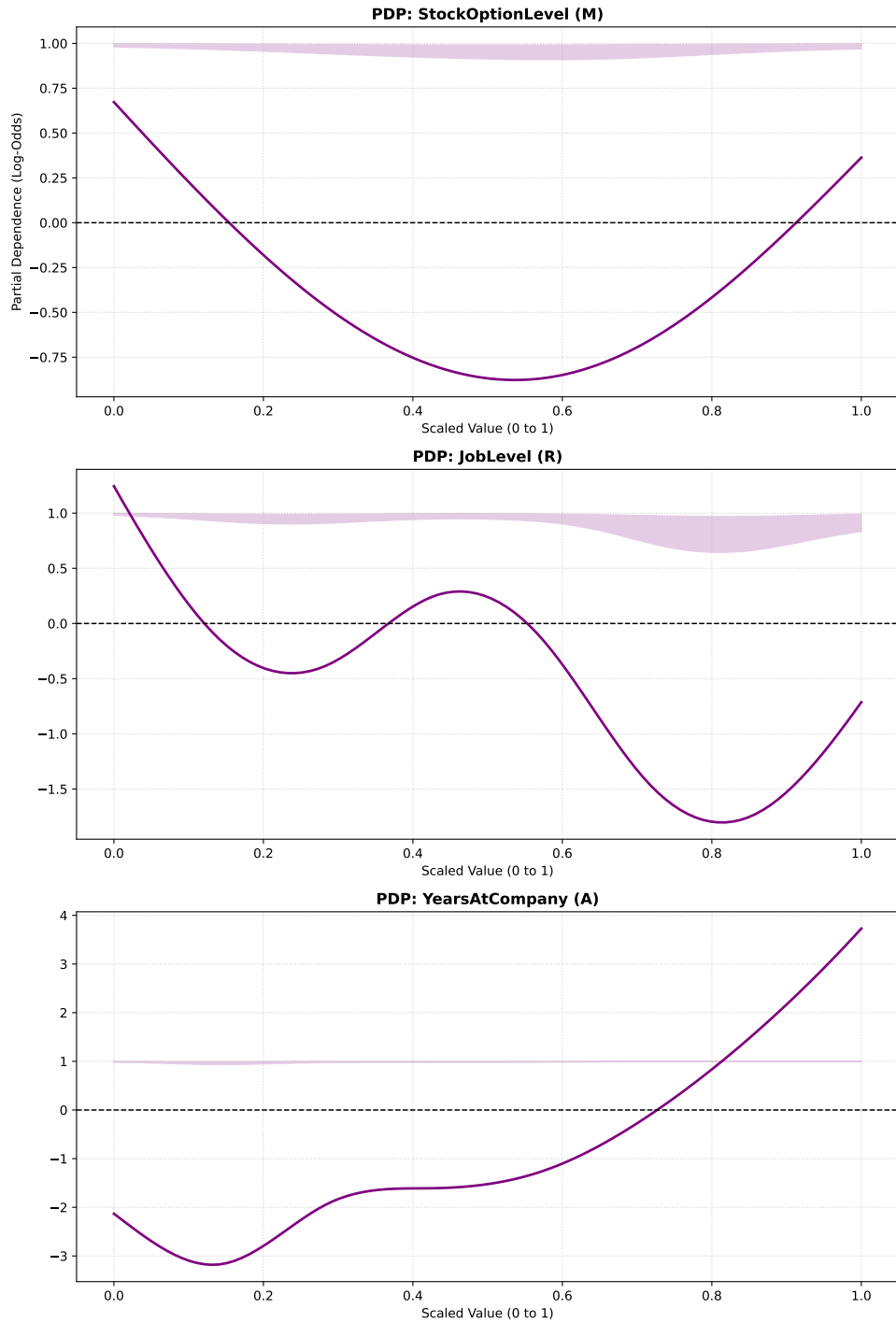


Figure 4: Non-Linear Feature Effects for Top Continuous Drivers (IBM GAM)

exponentially. Translated to Enterprise HR strategy, this insight informs us that *Institutional loyalty* has an expiration date in that employees may very well be *retained* comfortably by an enterprise for several years but if they arrive at the late stages of their *tenure* without having been promoted or structurally re-engaged, “*tenure fatigue*” is likely to take hold and exponentiate their *attrition risk*.

While the **PDP plots** from the **GAM model** offer tremendous theoretical clarity in the full spectrum observation of *workforce behavior*, their practical limitations in implementation (e.g., an HR manager cannot meaningfully deploy a “curve” into an automated alert system) necessitate the hard binary logic of **Decision Trees** to locate specific “*Kill Zones*” of *turnover* in an effort to bridge the gap between *behavioral theory* and operational HR intervention.

1.1.5 Decision Tree Attrition “Kill-Zones” (IBM Dataset)

Kill Zone 1: 92.8% Attrition Concentration

Impact: 0.9/1.0 weighted score (Applies to 35 actual employees | 2.80% of workforce)

Logic Path: `OverTime_0 (S) <= 0.50 AND JobLevel (R) <= 1.50 AND YearsInCurrentRole (R) <= 0.50`

Kill Zone 2: 90.6% Attrition Concentration

Impact: 0.9/1.0 weighted score (Applies to 20 actual employees | 1.60% of workforce)

Logic Path: `OverTime_0 (S) <= 0.50 AND JobLevel (R) <= 1.50 AND YearsInCurrentRole (R) > 0.50 AND DistanceFromHome (S) > 12.50`

Kill Zone 3: 89.4% Attrition Concentration

Impact: 0.9/1.0 weighted score (Applies to 21 actual employees | 1.68% of workforce)

Logic Path: `OverTime_0 (S) <= 0.50 AND JobLevel (R) > 1.50 AND Department_Sales (R) > 0.50 AND StockOptionLevel (M) <= 0.50 AND DistanceFromHome (S) > 5.50`

he cohort specified by Kill Zone 1 is comprised of Junior employees (`JobLevel <= 1.50`) brand new to their role (`YearsInCurrentRole <= 0.50`) who are recruited into working *overtime* (`OverTime_0 <= 0.50`, `OverTime_0` must be equal to 1 to indicate the absence of *overtime*). In an HR context this stipulates that if IBM hires new employees into brand new roles and forces them to work beyond business hours this will come at the cost of 93% percent of this cohort *attriting*, these employees would *burn out* before their *onboarding* is even completed in earnest.

Kill Zone 2 represents a class of more tenured Junior employees (`JobLevel (R) <= 1.50 AND YearsInCurrentRole (R) > 0.50`) than in Kill zone 1, like the employees in Kill Zone 1, these employees also work *overtime* (`OverTime_0 <= 0.50`) however they are uniquely beleaguered by long *commutes* to the office (`DistanceFromHome > 12.50`). The curative HR remedy to this issue would be the facilitation of additional *remote work* days or transportation stipends to alleviate the physical *exhaustion* faced by Junior employees who struggle with both extended working ours and arduous *commute* journeys.

Kill Zone 3 offers a unique look at the vulnerabilities of organizational *veterancy* particularly in *Sales roles* (`JobLevel (R) > 1.50 AND Department_Sales (R) > 0.50`), this cohort also suffers from compromised *work life balance* (`OverTime_0 (S) <= 0.50`) as well as the limitations of IBM's reward structure (`StockOptionLevel (M) <= 0.50`). This logical path points to classic *sales burnout* observed in enterprises all over the world. Experienced sales representatives are being compelled to grind beyond business hours without long-term *extrinsic rewards* to lock in their *commitment*. Absent this *compensatory equity*, veteran sales staff will simply take their client books to a competitor.

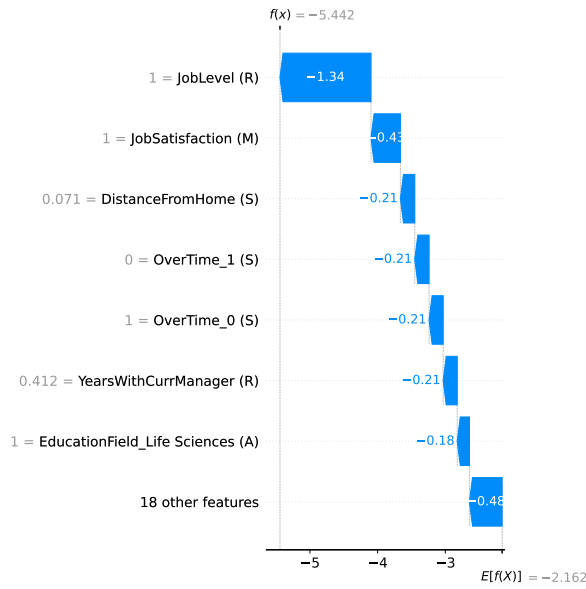
While **Decision Trees**, as illustrated by the 3 Kill-Zones, can produce watershed understandings of high-risk groups within an organization, HR departments ultimately concern themselves with the well-being of individuals. Managers need to know with specificity which of the employees they manage is a *flight risk*. To address this need, this study employs **SHAP Extreme Case Analysis** in the next section as the final proof of concept by demonstrating that the pipeline can surgically diagnose the distinct mathematical pressures acting on a single employee.

1.1.6 SHAP Extreme Case Analysis: The Safe Bets & *Flight Risks* at IBM

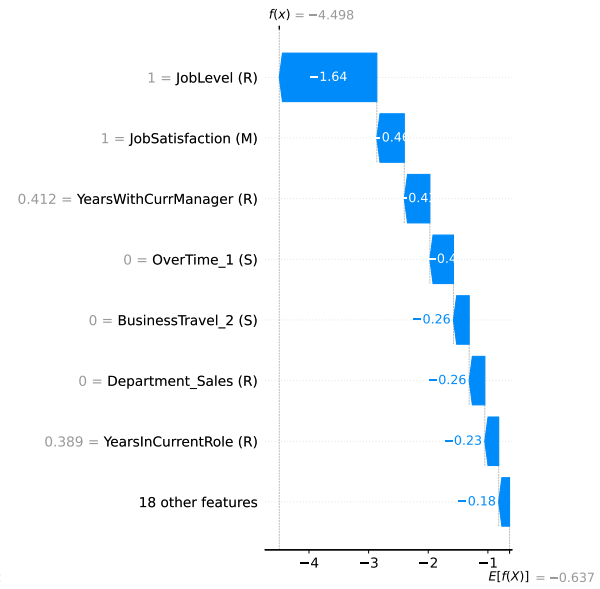
Below in Figure 5, we notice that the massive blue bars pushing the employee’s risk profile into the negative (safe/*retention*) territory are `JobLevel` and `YearsWithCurrManager`. The global insights observed in Section 1.1.3 are validated here at the individual level in our discovery that the most *retainable* people at IBM aren’t necessarily the most generously *compensated* but instead employees grounded in a deep structural *seniority* by virtue of their roles in the organization and also enjoy long standing positive *relationships with their managers*.

The riskiest employees as seen in Figure 6 are those plagued by forced *overtime* (`OverTime_1`) and critical of their environmental situation at the organization (`DistanceFromHome`). Herein lies the surgical power of **SHAP analysis** as it can be inferred from this diagnostic panel (Figure 6) that a successful *retention strategy* for these high-risk employees must manifest as schedule and *workload relief* as opposed to incremental *compensatory bonuses* like salary raises.

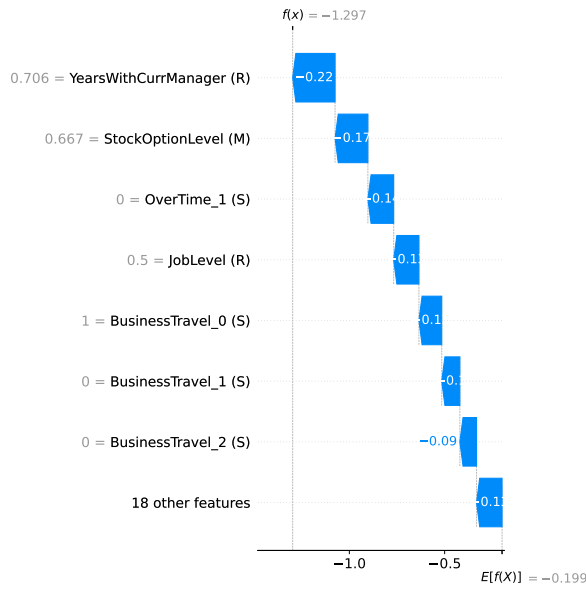
Our pipeline thus far as implemented in Section 1.1 successfully proves that highly accurate and individualistic diagnostics can be just as efficacious as those produced by “**black-box**” **models**. Hence, with the theoretical and diagnostic capabilities of the **models** proven, the final step in the IBM evaluation is an empirical ranking and tabulation to determine which **algorithm** actually caught the most *flight risks*.



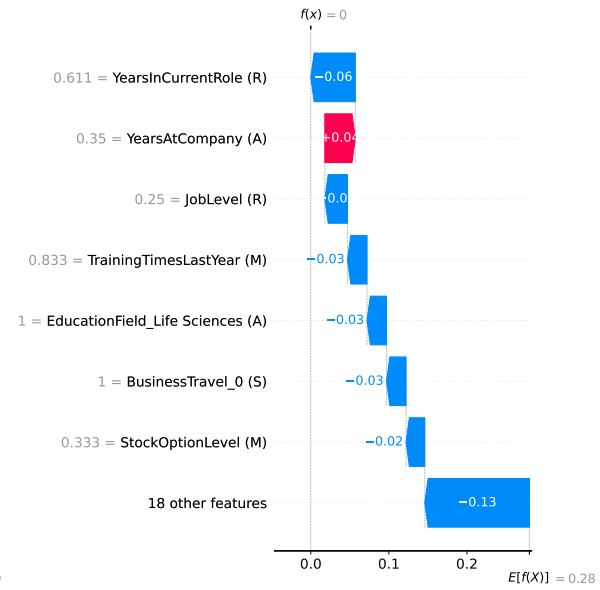
(a) A: Std Logistic Regression - Safest



(b) B: Reg. Logistic Regression - Safest

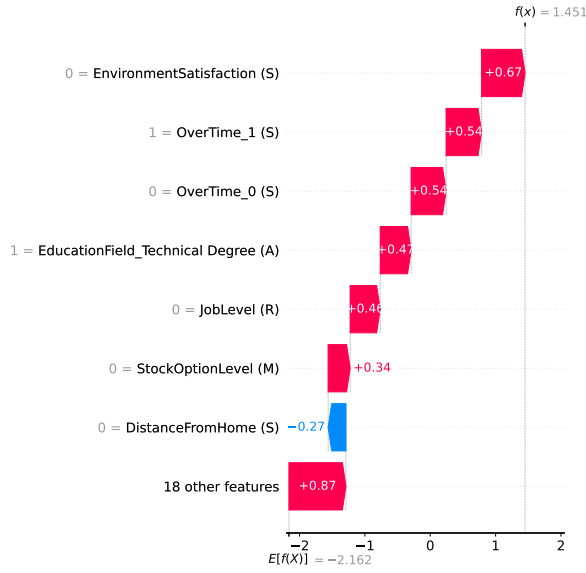


(c) C: Linear SVM (Tuned) - Safest

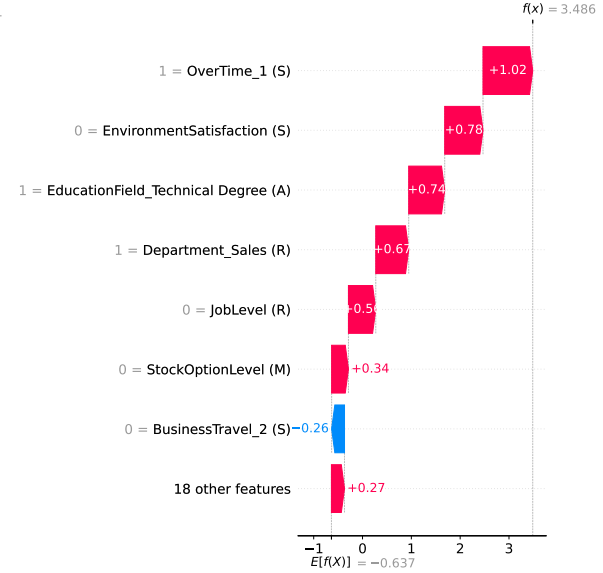


(d) F: GAM - Safest

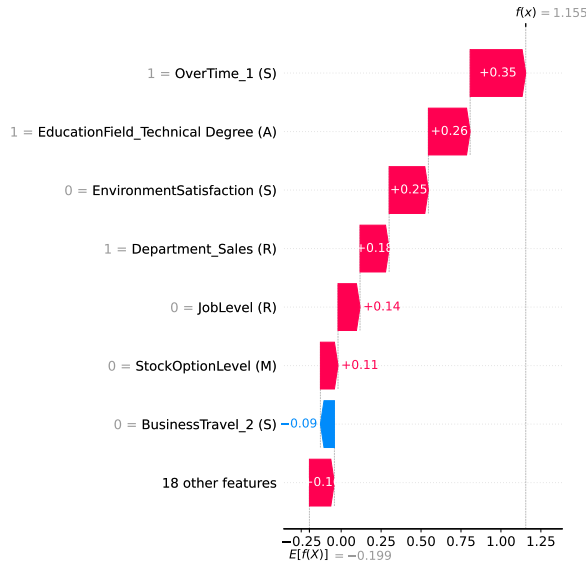
Figure 5: SHAP Local Interpretability: Profiles with Lowest Attrition Probability (IBM)



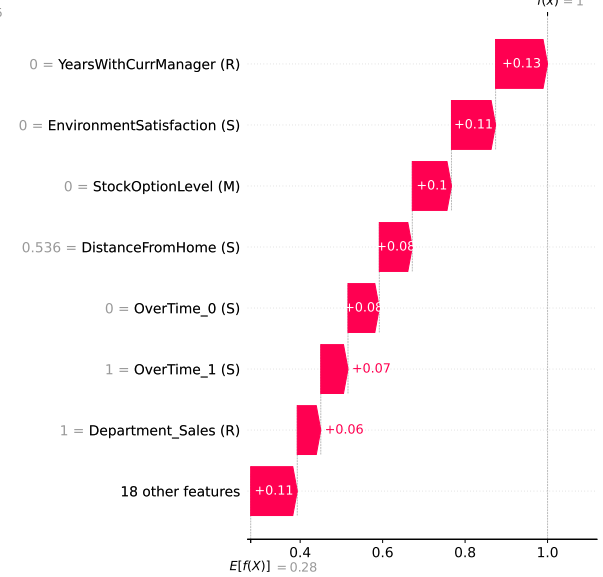
(a) A: Std Logistic Regression - Riskiest



(b) B: Reg. Logistic Regression - Riskiest



(c) C: Linear SVM (Tuned) - Riskiest



(d) F: GAM - Riskiest

Figure 6: SHAP Local Interpretability: Profiles with Highest Attrition Probability (IBM)

1.1.7 Model Performance Matrix (IBM Dataset)

Table 1: Consolidated Algorithm Performance and Hyperparameters (IBM Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1- Score (Weighted)
1	D: Decision Tree	{‘min_samples_split’: 2, ‘min_samples_leaf’: 20, ‘max_depth’: 5, ‘criterion’: ‘log_loss’}	0.681	0.75	0.731
2	F: Generalized Additive Model (GAM)	{‘lam’: 0.1}	0.795	0.722	0.804
3	C: Linear SVM (Tuned)	{‘C’: 0.0379269019073225}	0.741	0.694	0.764
3	B: Regularized Logistic Regression	{‘solver’: ‘saga’, ‘penalty’: ‘l2’, ‘C’: 555.556}	0.735	0.694	0.76
5	E: k-Nearest Neighbors	n_neighbors=7	0.957	0.417	0.857
6	A: Standard Logistic Regression	Baseline (Untuned)	0.989	0.222	0.83

We see in Table 1, the **Decision Tree** performs best with the primary **evaluation metric** of this study by securing an **Attrition Recall rate** of 0.75 implying that the **algorithm** successfully identifies 3 in 4 employees that actually terminated, this is in stark contrast to the untuned **baseline model (Model A: Std Logistic Regression)** which with an **Attrition recall** of 0.22 could only muster the successful identification of 1 in 5 *leavers* which cements **Model D** as a substantial computational upgrade to this baseline.

The best holistic compromise however is **Model F (Logistic GAM)** which earned a near comparable primary **metric score** of 0.722 to **Model D**’s 0.75, but maintained a much higher F_1 **score** than its competitor in **Model D** ($F_1 = 0.724$), this is due to its superior performance in the domain of **retention recall**. The enterprise HR implication of this tabulation therefore avers that while **Model D** flags the most *flight-risks*, **Model F** offers comparable performance for fewer tradeoffs as it avoids burying HR in false-positive *retention* meetings.

Hence Table [1](#) validates the introductory hypothesis of this study that the evolution of data pipelines from rigid, **lineal algorithms** (**Models A through C**) into inherently transparent and interpretable **non linear models** like **GAM** and **Decision trees** not only provided superior business intelligence but also yielded the highest absolute **predictive performance**.

1.2 Results: Babushkin Dataset

Because the Babushkin dataset was synthetically compiled as a **balanced dataset** (there is parity between records indicating retention and termination), the computed class weights are {Retention: 1.01, Attrition: 0.99} owing to the absence of a massive class imbalance in the dataset to skew the baseline.

The Babushkin dataset sees an overall increase in features in spite of dropping non-informative and demographic columns due to the *one-hot encoding technique* producing multiple binary variations of the categorical columns from the native unmodified dataset.

Babushkin Dataset Pipeline | Shape Filtered: (1129, 16) -> (1129, 22)
Computed Imbalance Weights: {0: 1.01, 1: 0.99}

Final MARS-Tagged Features:

- stag (A)
- greywage (S)
- extraversion (M)
- anxiety (M)
- selfcontrol (M)
- novator (M)
- independ (M)
- hr_profession (R)
- coach_my head (R)
- coach_no (R)
- coach_yes (R)
- way_bus (S)
- way_car (S)
- way_foot (S)
- traffic_Agency_Ads (S)
- traffic_Job_Boards (S)
- traffic_Referral_Network (S)
- industry_Heavy_Industry (R)
- industry_Public_Sector (R)
- industry_Retail_Services (R)
- industry_Tech_Professional (R)

1.2.1 kNN Elbow Diagram & Feature Selection (Babushkin Dataset)

We observe here in the tuning of the **kNN model** that the **ANOVA F-test** employed by the Python **SelectKBest** function has violently pruned the *Five-Factor Model (FFM) personality traits*. The features **anxiety (M)** (corresponds to *Neuroticism*) and **independ (M)**

(corresponds to *Openess to Experience*) were the only two factors of the *Five-Factor Model* that met the threshold of **linear dependency** with the target variable, this selection makes salient the limited utility of *extensive personality screening* as a procedural step in *enterprise hiring practices*.

kNN will now use only these 10 features:

```
['stag (A)' 'anxiety (M)' 'independ (M)' 'hr_profession (R)'
 'coach_yes (R)' 'way_bus (S)' 'way_foot (S)' 'traffic_Job_Boards (S)'
 'traffic_Referral_Network (S)' 'industry_Public_Sector (R)']
```

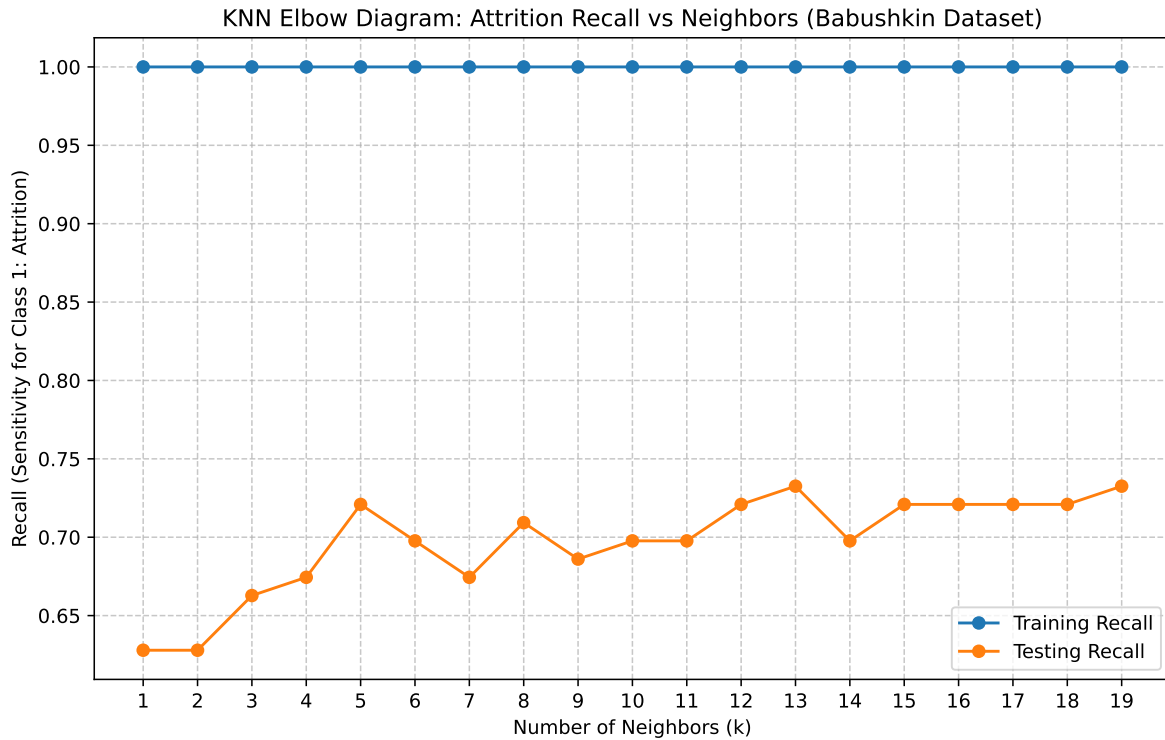


Figure 7: KNN Elbow Diagram: Attrition Recall vs. Neighbors (Babushkin Dataset)

Manually Selected for Interpretability: $k=5$ with Test Recall of 0.7209

Though the zenith of **recall** in Figure 7 is met only at $k = 13$ or higher, a manual override of $k = 5$ is made to ground the implementation of this algorithm in *business practicality*: While HR managers can meaningfully oversee and intervene in a *localized pod or team* of the employee's 5 closest peers, allowing the algorithm to optimize blindly and demand a *peer group* of 13 or more people would compromise its servicability as *peer groups* this large exceed *organizational spans of control* that most humans are capable of overseeing.

1.2.2 Global Feature Importances (VIMS Facet Grid - Babushkin Dataset)

Particularly arresting about Figure 8 above is the contrast between both **Logistic Regression** models (**Models A & B**), while **Model A** insists that **anxiety (M)** and **stag (A)**(tenure) are dominant in ascertaining *flight risk*, **Model B** favors Role based and situational factors like **hr_profession (R)** and **way_foot (S)**. This occurs due to the **L2 penalization** implemented by **Model B** which eradicates **anxiety (M)** and deprioritizes **stag (A)**. This informs us that the baseline model was taken in by correlated noise but by the day to day realities of their incumbent role (**hr_profession (R)**) and the logistical ease of their commute (walking to work acting as a massive *retention anchor* via the **way_foot (S)**)

A noticeable drop off in feature importance also presents itself in Figure 8 with **Model D (Decision Tree)**, while the **linear models** distribute importance across a wide curve, the **decision tree** restricts its prioritization to merely four variables with **stag (A)** topping the list followed distantly by **traffic_Job_Boards (S)**, **selfcontrol (M)**, and **independ (M)**. This in part due to the computational nature of **decision trees** in being liberated from charting gradual psychological probabilities or drawing straight lines. Metrics like **Gini impurity** or **log-loss** are instead operationalized to aggressively hunt for the single cleanest mathematical cut that divides stayers from leavers. This suggests that if HR managers are to fashion a hard *binary alert system* to flag turnover, they would be well advised to ignore psychological surveys entirely. **Model D** champions **stag (A)** as the most mathematically pure threshold to flag a *flight risk*, making almost all other features computationally irrelevant for the initial operational filter.

Gleaning at Figure 8 above in a more wholistic sense, an observable consensus between the **linear models** emerges that **anxiety (M)** is the most crucial *retention factor*, this congruence between models is credible due to varied geometric approaches each model takes en route to this conclusion: **Model A** uses **maximum likelihood estimation** to find a trend across the entire workforce. **Model C**, however, only cares about the **support vectors** (i.e, the specific employees sitting right on the mathematical boundary between staying and leaving).

In enterprise context this means **Model A** postulates that high anxiety serves as a *retention anchor* for the global workforce while **Model C** offers a more surgical conclusion: for the employees who face imminent attrition risk (the **support vectors** teetering on the edge of the *flight risk* boundary), their level of *risk aversion* (anxiety) is the definitive tie-breaker that keeps them in their seats.

Model E serves to marry some of the divergent perspectives (logistical and psychological) of the models discussed above by **way_foot (S)** , **hr_profession (R)** , and **anxiety (M)** as its top magnitude drivers which lends itself well to the idea that HR managers can seldom afford to overlook the impact of physical logistics (having to walk or take public transit to work) and base line *risk aversion* (i.e employees forming “*continuance committments*” with their employer due to latent or realized fears of job loss).

Global Feature Importance by Algorithm (Babushkin Dataset)
 Red = Attrition Driver | Blue = Retention Factor | Grey = Magnitude Only

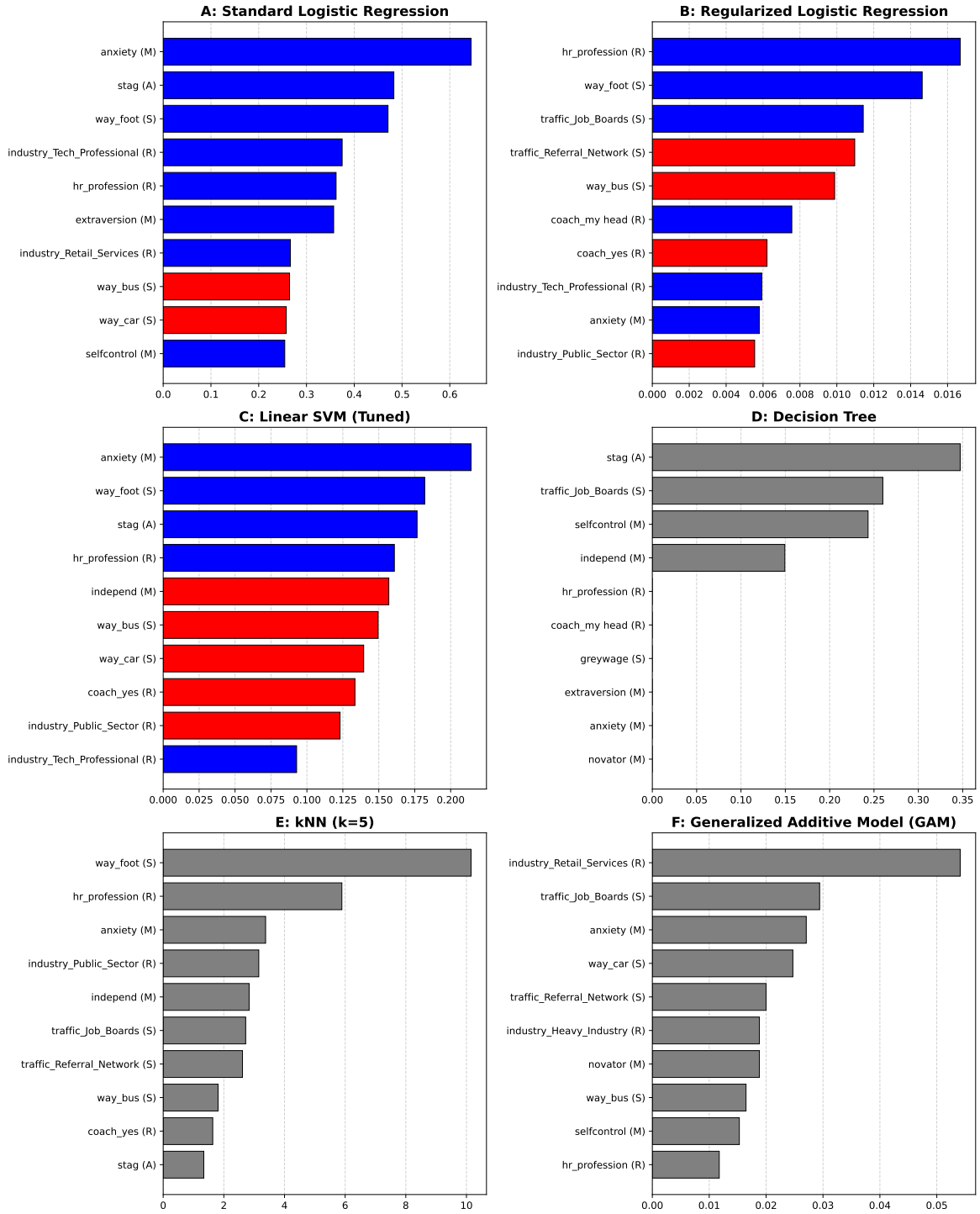


Figure 8: Global Feature Importance by Algorithm (Babushkin Dataset)

Finally, **Model F** violently deviates from any consensus observed in **Models A through E** in its elevation of **industry_Retail_Services (R)** and **traffic_Job_Boards (S)** to its absolute top spots and in doing so reveals that industry placement (specifically being in Retail) and sourcing origin (Job Boards) possess highly complex, non-linear relationships with turnover at Babushkin. Thereby **Model F** hints that the *attrition curves* for job board sourced retail workers distinguish themselves from their other enterprise counterparts.

1.2.3 Logistic GAM Partial Dependence Plots (Babushkin Dataset)

In this section, the exploration of binary dummy variables via **Partial Dependence Plots** are avoided due to the purpose of **PDP plots** being oriented towards visualizing the marginal effect of a feature across a continuous spectrum. As a result, when **PDP plots** are implemented on binary dummy variables like the **industry_Retail_Services (R)** flagged as a top *attrition/retention factor* by **Model F**, the results often reveal themselves to be disjointed step functions with minimal interpretive value. Therefore, in order to sufficiently channel the **spline** capabilities of the **GAM**, the pipeline intentionally bypassed those categorical variables to plot the top continuous drivers: **anxiety (M)**, **novator (M)**, and **selfcontrol (M)**

The top-most visual in Figure 9 which explores the marginal effects of the **anxiety (M)** feature points to employee anxiety being a crucial *retention anchor* for the Babushkin enterprise, this is visibly confirmed by observing that for employees with zero anxiety (0.0), the **log odds** of attrition sit dangerously high in the positive *flight risk* territory ($\sim +0.4$). However, as the anxiety scale advances to the **X-axis** upper limit of 1.0, the curve plunges cleanly across the **zero-line**, bottoming out at roughly -0.4 and thus furnishes a counter-intuitive behavioral reality in that highly anxious employees are massive *retention anchors*, this also suggests a willingness among *disenamored employees* of the Babushkin organization to retain employment rather than take on a perceived gamble in the wider job market.

The **PDP plot** for **novator (M)** is telling even in demonstrating zero **vertical variance** which signals that being an “innovator” has absolutely zero isolated predictive power regarding an employee’s *flight risk* and that initiative and creativity alone cannot translate to *longevity* at the Babushkin organization.

The middle plot in Figure 9 speaks to the virtues of restraint within Babushkin’s workforce. This is because within the **PDP plot** for **selfcontrol (M)**, the **log-odds** curve hovers in the slightly positive/neutral zone for employees with low to moderate self-control (0.0 to ~ 0.5). But once the feature crosses the 0.6 threshold, the **spline** violently dips into the negative *safe zone*. This points to employees with high degrees of self control also having the *self regulative ability* to side step *emotionally driven impulsive decisions* to exit the enterprise.

Non-Linear Feature Effects: Top Continuous Drivers (Babushkin GAM)

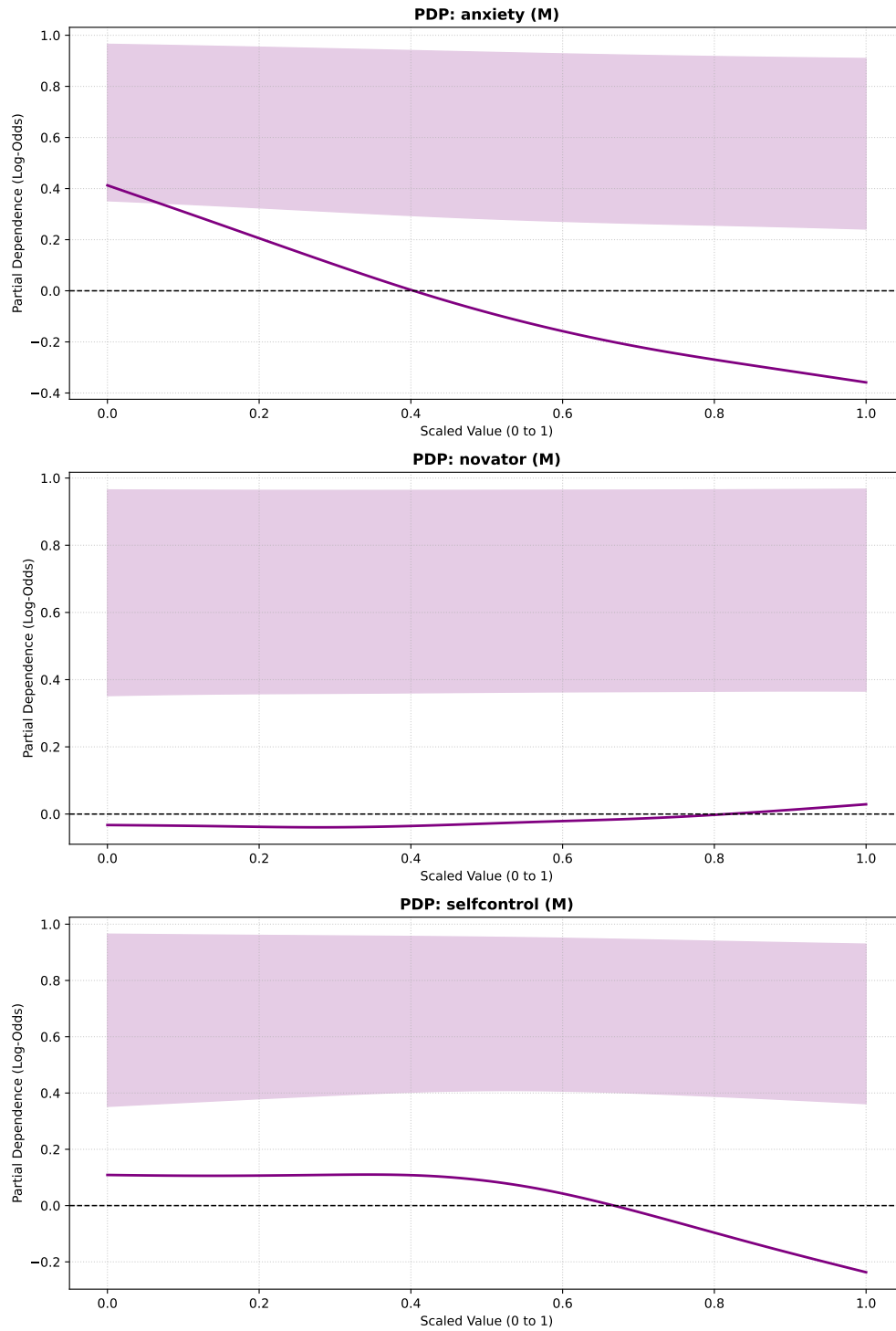


Figure 9: Non-Linear Feature Effects for Top Continuous Drivers (Babushkin GAM)

1.2.4 Decision Tree Attrition “Kill-Zones” (Babushkin Dataset)

Kill Zone 1: 63.6% Attrition Concentration

Impact: 0.6/1.0 weighted score (Applies to 261 actual employees | 27.22% of workforce)

Logic Path: stag (A) > 2.73 AND stag (A) <= 27.09 AND independ (M) > 5.15

Kill Zone 2: 55.1% Attrition Concentration

Impact: 0.6/1.0 weighted score (Applies to 256 actual employees | 26.69% of workforce)

Logic Path: stag (A) > 2.73 AND stag (A) > 27.09 AND traffic_Job_Boards (S) <= 0.50

Kill Zone 3: 48.7% Attrition Concentration

Impact: 0.5/1.0 weighted score (Applies to 193 actual employees | 20.13% of workforce)

Logic Path: stag (A) > 2.73 AND stag (A) <= 27.09 AND independ (M) <= 5.15

The **Kill zones** above readily reveal the expansive organization wide scope of *turnover* issues at the Babushkin workforce as each of these zones apply to anywhere from 20 to 27 % of the overall work force. This elevated scope suggests that *turnover* at Babushkin may well be more systemic and cultural than initially imagined by the other **predictive models** in past sections.

Kill Zone 1 represents a mid-tenured group (**stag (A)** (*tenure*) between 2.73 and 27.09 months) of self motivated individuals (**independ (M)** > 5.15) and possesses the highest **attrition concentration** at 63.6%, this might serve to guide the Babushkin organization’s HR managers in understanding that though employees might’ve survived *on boarding* and the *learning curve* of their roles, their self motivated natures also empower them to aggressively secure better offers elsewhere if dissatisfied with their employment, this is ostensibly due to the reality that by 27 months of *tenure* in the organizations, most individuals will not have been retained long enough to be locked in by deep *seniority* or *institutional inertia*.

Kill Zone 2 explores an identical *tenure group* as sequestered by **Kill Zone 1** but with *job sourcing* being the key differentiator of indicative *turnover intent*, since this group captures over 55% **attrition**, it signals to *HR leaders* that *enterprise talent sourcing strategies* cannot be understated in broader organizational efforts to stem *turnover*.

Kill Zone 3 finds itself as the third rung in a consensus between **decision tree paths** that mid-tenured groups at the Babushkin Organization are particularly beleaguered. This is because all 3 **Kill zones** confront different aspects of the exact same *tenure cohort* (**stag (A)** (*tenure*) between 2.73 and 27.09 months), with **Kill Zone 3** uniquely focusing on employees most in need of *managerial guidance* (**independ (M)** < 5.15).

These observations taken altogether in tandem (from **Kill Zones 1 through 3**) inform us that the mid-tenure demographic is the single most vulnerable demographic in the Babushkin organization representing over 50% the workforce and that available remedial steps *enterprise*

HR could take on include paying sufficient mind to which streams employees are being hired from and the implementation of a dual pronged strategy for the *retention* of mid tenured individuals: for more independent employees (**independ (M)** > 5.15 seen in **Kill Zone 1**) managers must offer extreme *autonomy*, new challenges, and clear *upward mobility* so that *task boredom* does not give way to *professional listlessness* and for more dependent employees (**independ (M)** < 5.15 seen in **Kill Zone 3**), managers must provide heavy *mentorship*, *structural scaffolding*, and *continuous feedback* in order to avoiding burning out or overwhelming these employees.

1.2.5 SHAP Extreme Case Analysis: The Safe Bets & *Flight Risks* at Babushkin

Across the diagnostic panel of most *retainable employees* in the Babushkin organization (Figure 10) the **models** paneled here near unanimously agree on the profile of the most stable employees (1 = **way_foot** (S) and 1 = **industry_Tech_Professional** (R)) by categorizing employees who work in the Tech division and physically walk to work as being most insulated from *attrition risk*. The impact of *direct, active mentorship* from an employee's supervisor is also a crucial driver for *retention* as seen by the prominent placement of the (1 = **coach_my head** (R)) variable in Figure 10, this finding is congruent with the canonical belief in HR circles that *adversarial relationships with direct supervisors* predict future *turnover events*.

When individualistically assessing the risks at the Babushkin organization via Figure 11, it is apparent that yet another consensus emerges between the **models** about the profiles of the hardest employees to retain. Here the red bars driving the risk are (0.096 = **anxiety** (M)) and (0 = **hr_profession** (R)), in an HR context, the highest *flight risks* at Babushkin are employees outside the HR department who possess virtually zero *baseline anxiety*. Their fortitude empowers them with the *psychological confidence* to test the *open job market* and their departmental placement outside HR may well also point to a general dissatisfaction with the opacity of *enterprise workforce management decisions*.

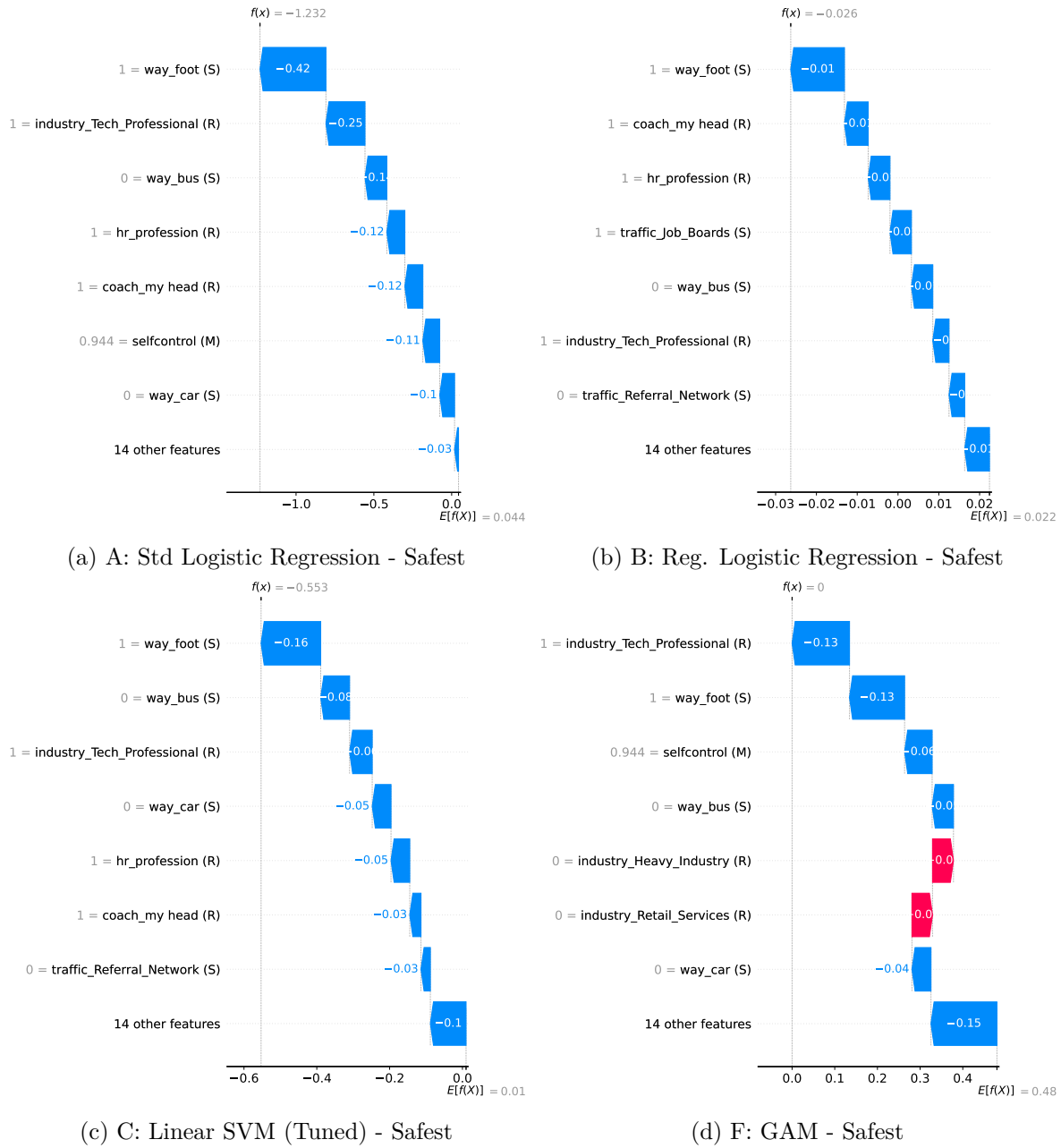
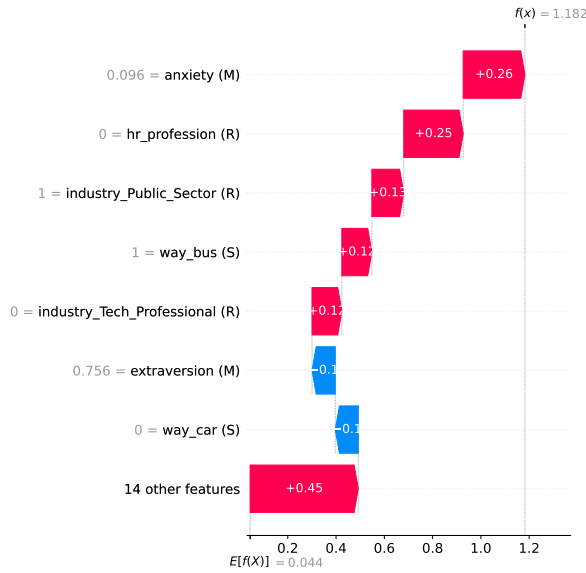
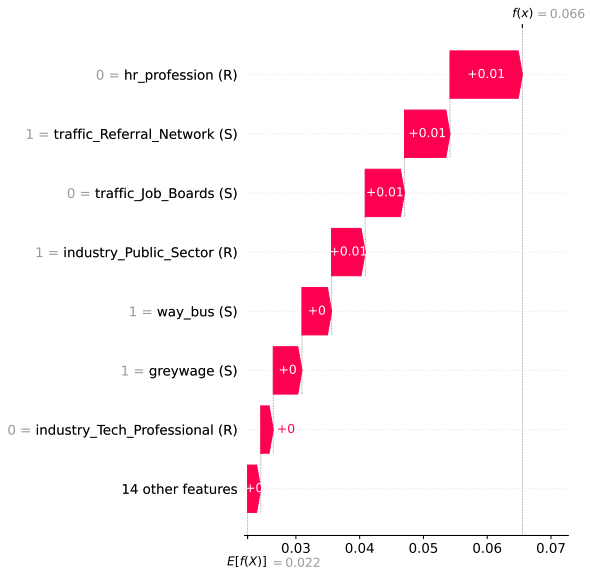


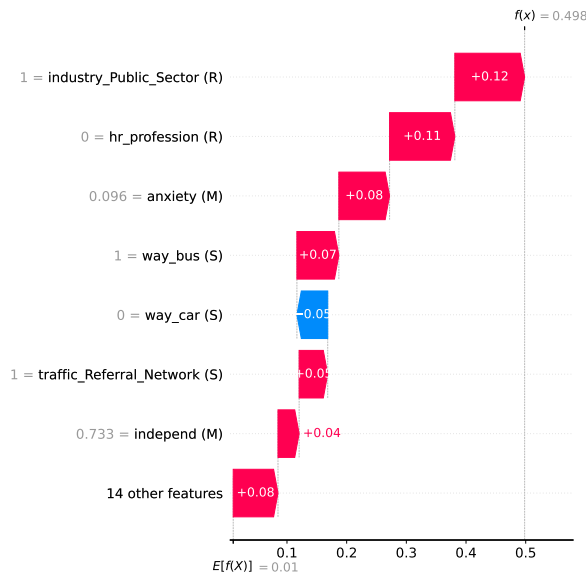
Figure 10: SHAP Local Interpretability: Profiles with Lowest Attrition Probability (Babushkin)



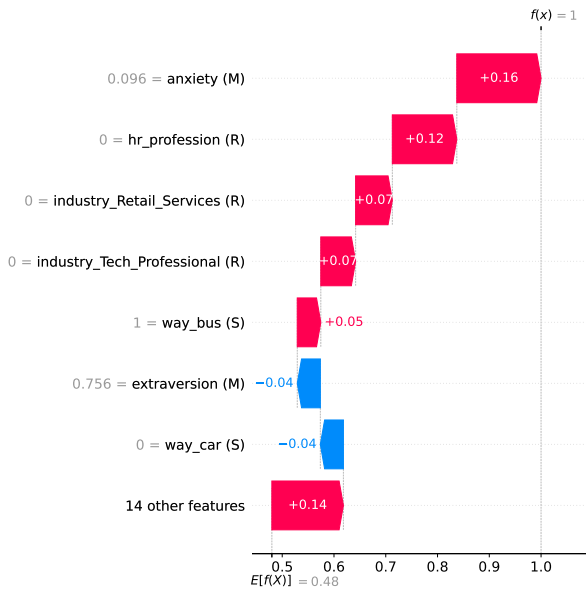
(a) A: Std Logistic Regression - Riskiest



(b) B: Reg. Logistic Regression - Riskiest



(c) C: Linear SVM (Tuned) - Riskiest



(d) F: GAM - Riskiest

Figure 11: SHAP Local Interpretability: Profiles with Highest Attrition Probability (Babushkin)

1.2.6 Model Performance Matrix (Babushkin Dataset)

The **performance matrix** in Table 2 below readily presents itself with a potential pitfall in its highest ranked model (**Model B**). Securing an **Attrition recall** of 0.953 **Model B** seemingly demonstrates a near perfect capability to predict *turnover events*, however as is well understood in the tuning of **turnover classification models** the maximization of **Attrition recall** can only be realized with the sacrifice of **retention recall** as evidenced by **Model B**'s retention **recall rate** which stands under 0.1. In practice, the implementation of **Model B** to flag attrition would lead to the squandering of *managerial bandwidth* and resources on *false-positive retention meetings* and related interventions.

This disqualification of **Model B** grounds **Model E (kNN with k = 5)** as the true victor of the contest between **classifier models** in its securing of a highly respectable **Attrition Recall** of 0.721 while maintaining a balanced **Retention Recall** of 0.583. Therefore by utilizing localized and inherently manageable *peer groups* (k=5), **Model E** successfully flags nearly 3 out of 4 actual *flight risks* without the additional risk of profligate *turnover intervention* and thus yields the highest overall **F1-Score** (0.651) of all the tuned **models**.

While an observer of Table 2 may at once notice across the board that F_1 **scores** that top out at 0.6 might be low when held against the **classification metrics** produced by sterile financial models or clinical data, it should be noted that the Babushkin dataset was already perfectly balanced at the onset as mentioned in the introduction of Section 1.2, these **models** were therefore stripped of their ability to artificially inflate their **accuracy** or **F1-Scores** by simply blindly guessing the **majority class** (employees who end up staying). A behavioral explanation for this tendency may lie in the Babushkin dataset's characteristic reliance on psychological traits gathered from self reported survey data which tend to suffer from noisiness and subjectivity as well as bear the risk of potentially dishonest reporting from survey respondents.

Table 2: Consolidated Algorithm Performance and Hyperparameters (Babushkin Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1- Score (Weighted)
1	B: Regularized Logistic Regression	{‘solver’: ‘saga’, ‘penalty’: ‘l2’, ‘C’: 0.001}	0.071	0.953	0.4
2	E: k-Nearest Neighbors	n_neighbors=5	0.583	0.721	0.651
3	F: Generalized Additive Model (GAM)	{‘lam’: 100.0}	0.56	0.651	0.605

Table 2: Consolidated Algorithm Performance and Hyperparameters (Babushkin Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1- Score (Weighted)
3	A: Standard Logistic Regression	Baseline (Untuned)	0.56	0.651	0.605
5	C: Linear SVM (Tuned)	{‘C’: 0.0379269019073225}	0.583	0.616	0.6
6	D: Decision Tree	{‘min_samples_split’: 6, ‘min_samples_leaf’: 20, ‘max_depth’: 3, ‘criterion’: ‘log_loss’}	0.476	0.558	0.517

1.3 Results: Data Science Dataset

As discussed in the Methods section of this study, the Data Science Job Change Dataset is large and imbalanced dataset, therefore the class weights rendered below will be used in the underlying hyperparameter tuning process to robustly penalize classification models for missing the minority class (attrition) in the target variable.

Additionally, in the ETL (extract, transform, load) phase of computation, the columns **experience** and **last_new_job** columns contained string artifacts (like <1, >20, and **never**) that were coerced into continuous numeric variables by refactoring via Python dictionaries. Furthermore, in the vein of data preservation, missing or null valued categorical data was not dropped and instead imputed with a constant “Unknown” value before one-hot encoding.

DS Job Change Dataset Pipeline | Shape Filtered: (19158, 14) -> (19158, 27)
Computed Imbalance Weights: {0: 0.67, 1: 2.01}

Final MARS-Tagged Features:

- last_new_job (M)
- experience (A)
- training_hours (R)
- city_development_index (S)
- enrolled_university_Full time course (M)
- enrolled_university_Part time course (M)
- enrolled_university_Unknown (M)
- enrolled_university_no_enrollment (M)
- relevent_experience_Has relevent experience (A)
- relevent_experience_No relevent experience (A)
- company_size_10-100 (R)
- company_size_100-500 (R)
- company_size_1000-4999 (R)
- company_size_10000+ (R)
- company_size_50-99 (R)
- company_size_500-999 (R)
- company_size_5000-9999 (R)
- company_size_<10 (R)
- company_size_Unknown (R)
- company_type_Early Stage Startup (R)
- company_type_Funded Startup (R)
- company_type_NGO (R)
- company_type_Other (R)
- company_type_Public Sector (R)
- company_type_Pvt Ltd (R)
- company_type_Unknown (R)

did not converge

1.3.1 kNN Elbow Diagram & Feature Selection (Data Science Dataset)

As was the case with the IBM and Babushkin datasets in Section 1.1.2 and Section 1.2.1 respectively, Python's Select K-best feature will be used to prune the large feature space down to the most influential 10 variables.

kNN will now use only these 10 features:

```
['last_new_job (M)' 'experience (A)' 'city_development_index (S)'  
'enrolled_university_Full time course (M)'  
'enrolled_university_no_enrollment (M)'  
'relevent_experience_Has relevent experience (A)'  
'relevent_experience_No relevent experience (A)'  
'company_size_Unknown (R)' 'company_type_Pvt Ltd (R)'  
'company_type_Unknown (R)']
```

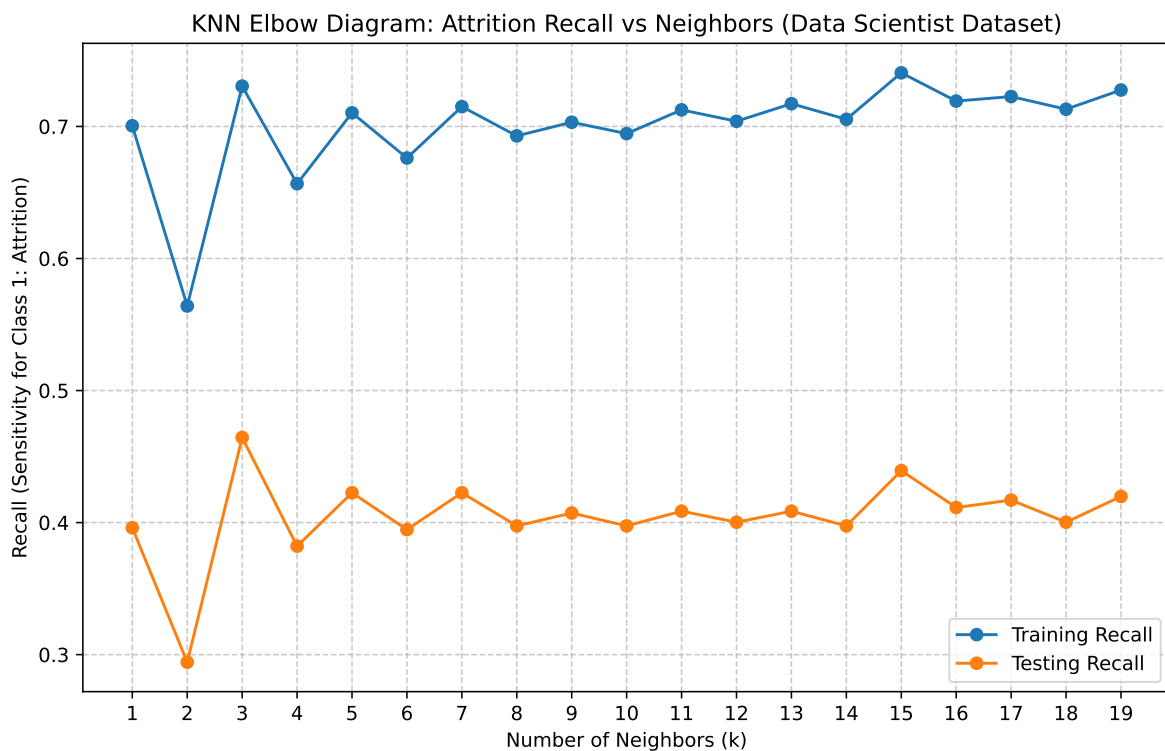


Figure 12: KNN Elbow Diagram: Attrition Recall vs. Neighbors (Data Science Dataset)

Peak Performance: $k=3$ with Test Recall of 0.4644

Figure 12 above peaks at $k = 3$, this is consequential in light of the optimal **k-values** produced by the other two datasets in this study (**k=7** for IBM and **k=5** for Babushkin). This hyper-localized **k-value** is likely a direct consequence of the dataset’s massive footprint (over 19,000 records) which serves to liberate the **algorithm** from needing to cast a wide geometric net to find statistical peers. It can locate near-exact “clones” within a very tight radius.

The translation of this finding to *Enterprise HR realities* suggest that in massive enterprises (such as Global market leaders and the like), HR need not confine itself to broad *departmental averages* to predict *data scientist turnover*. Large *talent pools* permit the **algorithm** to surgically flag *flight risks* based on the historical behavior of just a hand full of perfectly matched peers, allowing for highly individualized *interventions*.

1.3.2 Global Feature Importances (VIMS Facet Grid - Data Science Dataset)

Figure 13 shows us that the `city_development_index` (S) feature achieves something uniquely rare in this study: absolute universal dominance as it is the foremost predictive feature among the entire panel of assembled **models** and in doing so completely overrides the diverging consensus seen in previous sections between **linear and non-linear models**.

In a real word context, while it was observed in Section 1.2 and Section 1.1 that internal *departmental metrics* such as overtime and bad managers drove turnover, *macro-economic gravity* is the lynchpin of *retention* among *Data Science Professionals*. This likely stems from the nature of the industry in sourcing *talent* with highly mobile and premium skillsets and can result in *location hopping* as *Data Science professionals* mature in their careers. This is observable in the panels for **Models A through C** in Figure 13 called out by large **blue bars** which imply that if an employee is situated in a highly developed city or tech hub they tend to remain, however if located in a developing or lower tier market they are more likely to translate their experience and acumen to roles located in more economically favorable geographies.

Another striking element of Figure 13 is the placement of a synthetically created category (`company_size_Unknown` (R)) as the second most powerful **predictive feature** in modeling *attrition*, once again across the entire panel of **predictive models**. This validates the earlier methodological decision to retain **null valued categorical data**. A possible reason for this is when translated to the realities of *workforce dynamics*, missing data is rarely actually random. If a *data scientist* is recorded as working for an enterprise of unknown or unrecorded size, it strongly implies their engagement in a highly volatile *early-stage startup*, a *shell company*, or an unstructured *gig-economy contract* such as free-lancing.

Furthermore, **Models A, B & C** state that having relevant experience is a **predictor** of *attrition* (indicated by **red bars** for the `relevant_experience_Has relevant experience` (A) feature). This speaks to the existence of a favorable and lucrative *job market* for mid-level or experienced *data science professionals* and consequently *HR departments* must be cognizant of *poaching risk* as they train and mature their *data science teams*.

Global Feature Importance by Algorithm (DS_Job Dataset)
 Red = Attrition Driver | Blue = Retention Factor | Grey = Magnitude Only

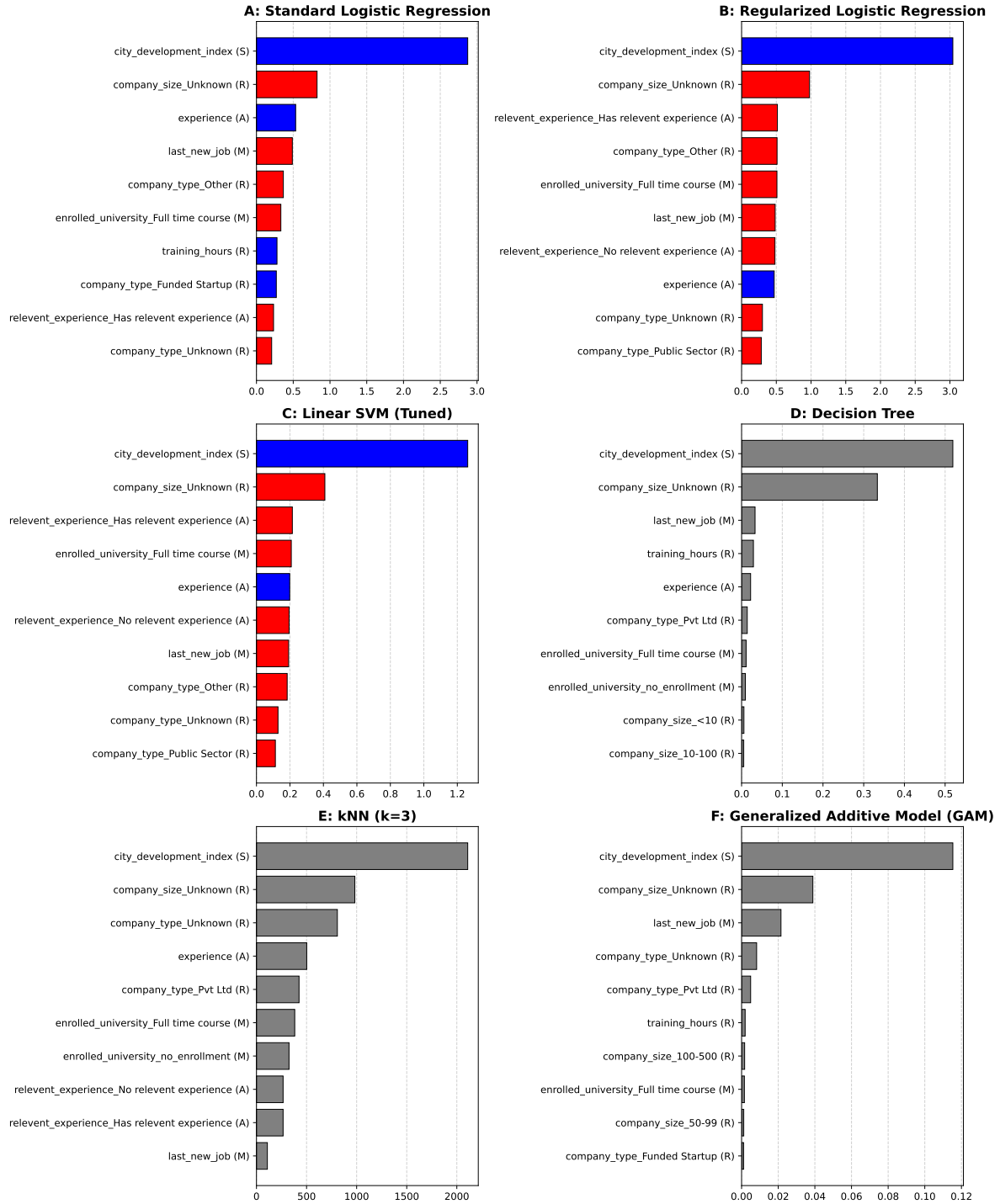


Figure 13: Global Feature Importance by Algorithm (Data Science Dataset)

1.3.3 Logistic GAM Partial Dependence Plots (Data Science Dataset)

Observable at once from Figure 14 is that the **PDP plot** for the **city_development_index (S)** feature is that it does not trend downward in a straight line and instead violently spikes of a massive *flight risk* peak at around the 0.3 scale mark, The 0.3 mark can be understood to be *developing or mid-tier economic markets* and *Data Scientists* stationed in these areas pose extreme *flight risk* (well punctuated by the plot peaking at a **log odds** of 2.0 implying an 88% probability of *imminent turnover*). It can be inferred that *Data Scientists* are likely using these roles to embellish their professional resumes before parlaying them into offers at more attractive geographic market (represented by the deep plunge into the negative/*retention zone* at the 0.8 mark and beyond), which instructs us that *enterprise geography* is a revolving door for premium *talent*.

The **PDP plot** for the **last_new_job** reveals the potential for *Infrastructural Traps* in the training of *data scientists*. This is visually observable on the plot as risk is deeply negative (safe) at the absolute 0.0 mark but climbs rapidly into the positive *flight-risk zone* and stays there. Green *data scientists* (where **last_new_job (M) = 0**) are temporarily anchored because they require massive *organizational infrastructure* such as *cloud computing subscriptions*, *distributed architecture*, and live *databases* to actually mature their skills in contemporary practice. They are thus occupationally and technologically precluded from leaving till they leverage the enterprise to build competencies in *Cloud computing* (like AWS or Azure), once this leverage is secured the *retention anchor* vanishes, and they immediately shop those subsidized skills to competitors.

Similarly, the plot for the **training_hours** exposes a ‘sweet-spot’ for training with an distinguishable nadir (its strongest *retention power*) around the 0.7 scaled mark which it breaks into after elevated *flight risk* seen at lower levels of training. After this localized dip the curve spikes back up toward the 0.9 and 1.0 marks. This indicates that For *data science professionals*, training is not just *corporate onboarding*; it is vital *technological upskilling*. Providing a substantial, heavy investment in their training (the 0.7 mark) secures deep loyalty because the enterprise is actively funding their *human capital*. However this investment poses risk of diminishing returns at the highest levels of training as if developing employees are starved of project work they are consequently starved of *psychological ownership*. The ensuing *task boredom* provides the impetus to direct the company’s training resources to certify themselves for a job elsewhere.

Non-Linear Feature Effects: Top Continuous Drivers (DS GAM)

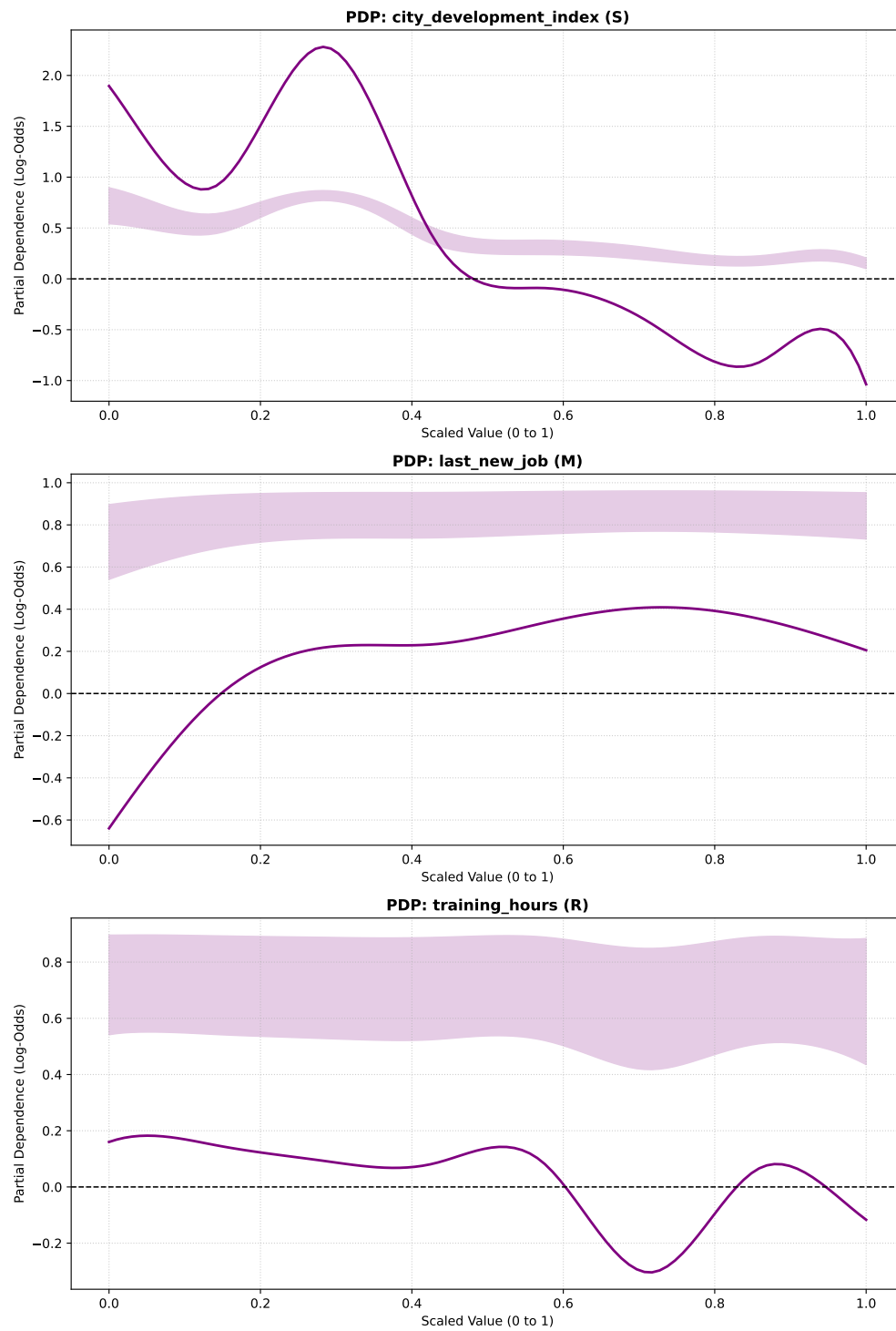


Figure 14: Non-Linear Feature Effects for Top Continuous Drivers (Data Science GAM)

1.3.4 Decision Tree Attrition “Kill-Zones” (Data Science Dataset)

Kill Zone 1: 93.8% Attrition Concentration

Impact: 0.9/1.0 weighted score (Applies to 30 actual employees | 0.18% of workforce)
Logic Path: `city_development_index (S) <= 0.62 AND experience (A) <= 1.50 AND
relevent_experience_No relevent experience (A) <= 0.50 AND training_hours (R) > 19.50
AND training_hours (R) > 47.50 AND training_hours (R) <= 86.00`

Kill Zone 2: 92.9% Attrition Concentration

Impact: 0.9/1.0 weighted score (Applies to 43 actual employees | 0.26% of workforce)
Logic Path: `city_development_index (S) <= 0.62 AND experience (A) > 1.50 AND
training_hours (R) <= 73.50 AND company_size_10-100 (R) > 0.50 AND training_hours (R)
> 43.00`

Kill Zone 3: 92.6% Attrition Concentration

Impact: 0.9/1.0 weighted score (Applies to 36 actual employees | 0.22% of workforce)
Logic Path: `city_development_index (S) <= 0.62 AND experience (A) <= 1.50 AND
relevent_experience_No relevent experience (A) <= 0.50 AND training_hours (R) <=
19.50`

All three **killzones** in Section 1.3.4 are brought alive by a striking commonality in having the exact same **root node**: `city_development_index (S) <= 0.62`. The constructed **decision tree** therefore affirms our earlier inclination from Section 1.3.3 that *Data scientists* stationed in a *developing or lower-tier economic market* are inherently at *risk* thereby grounding the primacy of *geographic reality* in determining *turnover risk* in this industry.

Also revealed in **Kill Zones 1 to 3** are the surgical lethality of *attrition risk* with respect to the *Attrition risk zones* seen in earlier sections like Section 1.2.4 which encapsulated 20-27% of the entire workforce. Here, the Data Science **Kill Zones** are applicale to tiny **micro-cohorts (0.18% to 0.26% of the workforce)**. This indicates that *sectoral turnover* in the Data Science space is not the *company-wide cultural rot* like it was at the Babushkin enterprise but instead localized hyper-specific, and extraordinarily lethal (with **attrition concentrations sitting at a staggering 92.6% to 93.8%**) all but guaraunteeing *turnover* if these specific conditions are met.

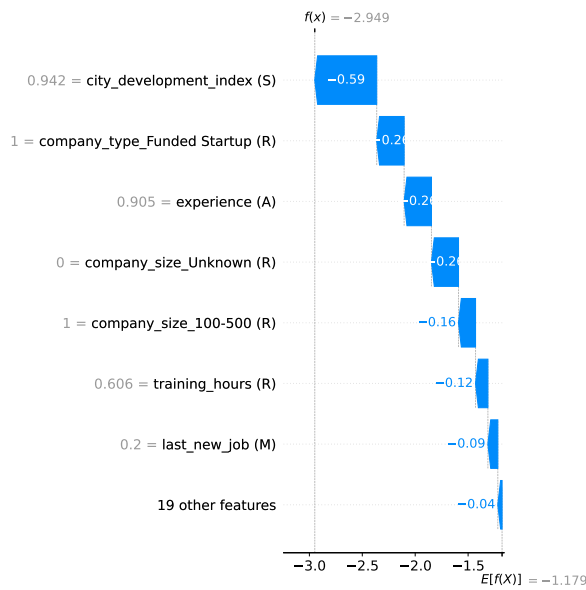
We also see divergent groups of *at-risk junior employees* with **Kill Zones 1 and 3**, here both **Kill zones** target a more seasoned cadre *Junior employees (experience <= 1.50)* who already possess *relevant industry background (relevent_experience_No relevent experience <= 0.50)* with **training hours** serving to distinguish one cohort from the other as **Kill Zone 1 (93.8% attrition)** captures juniors receiving moderate/heavy training (**between 47.5 and 86 hours**)and conversely **Kill Zone 3 (92.6% attrition)** captures juniors receiving almost no training (**training_hours <= 19.50**). This speaks to the susceptibility of *training programs*

in being used as *launch pads* for *voluntary turnover* in the case of **Kill Zone 1** and to *stagnant career prospects* being the *attrition driver* for **Kill zone 3**.

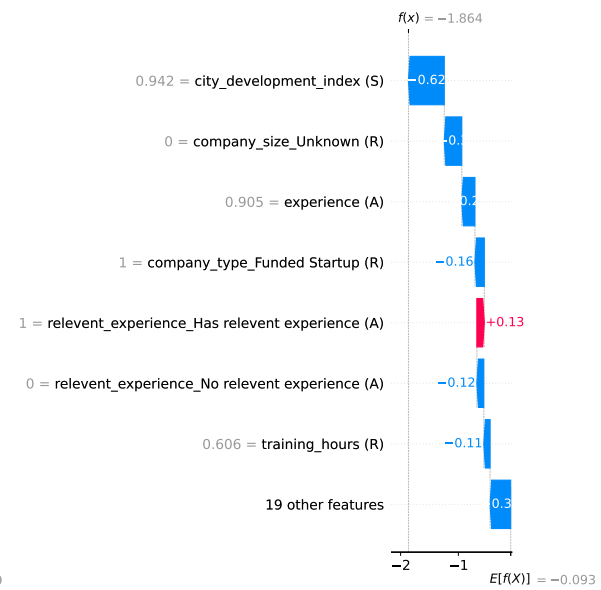
There are noticable glimpses of the “*Big Fish, Small Pond*” Syndrome presented by **Kill Zone 2 (92.9% attrition)** which targets more *experienced professionals* (**experience > 1.50**) in *developing cities* (**city_development_index <= 0.62**) working at *small firms* (**company_size_10-100 > 0.50**) with moderate training (**43 to 73.5 hours**). It is thus apparent that highly *experienced data scientists* stationed in *small companies* within *mid-tier cities* quickly hit a *professional ceiling*. Sans the massive *computational infrastructure* of a larger enterprise their careers may be destined to outgrow their environment and inevitably defect to *market competitors*.

1.3.5 SHAP Extreme Case Analysis: The Safe Bets & *Flight Risks* for the Data Science dataset

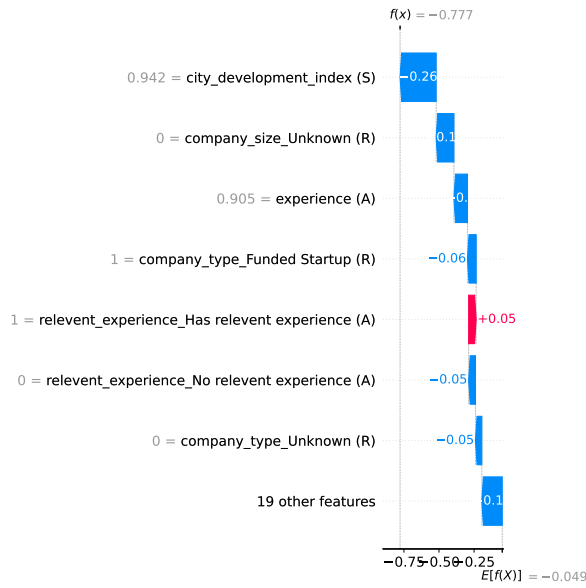
The panel of safest employees produced by Figure 15 validates the *geographical baseline* established in Section 1.3.4 and Section 1.3.3 as the prime driver of *retention* among *Data Science professionals*. As it pertains to *turnover* a noticable contrast also presents itself in these plots between the allure of *agile capital* on the one hand (inuniciated by **1 = company_type_Funded Startup (R)** as a major **blue retention anchor**) and the aversion to *bureaucracy* on the other (seen as **1 = company_type_Public Sector (R)** as a **red attrition driver** in Figure 16). This speaks to a trend of *data science profesionales* demanding agility backed by capital. They are highly retained by *well-funded startups* where they can build cutting-edge tech but chafe under the pressures endemic to the slow-moving, bureaucratic constraints of the *Public Sector* such as municipal budgets and the reliance of *public sector enterprises* on aging legacy tech stacks.



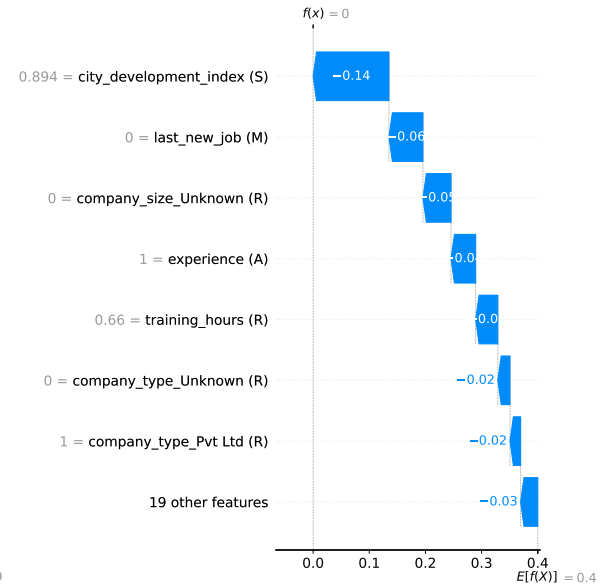
(a) A: Std Logistic Regression - Safest



(b) B: Reg. Logistic Regression - Safest



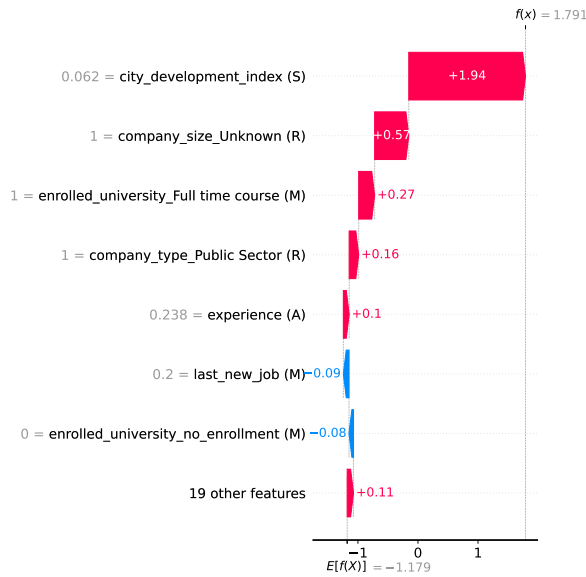
(c) C: Linear SVM (Tuned) - Safest



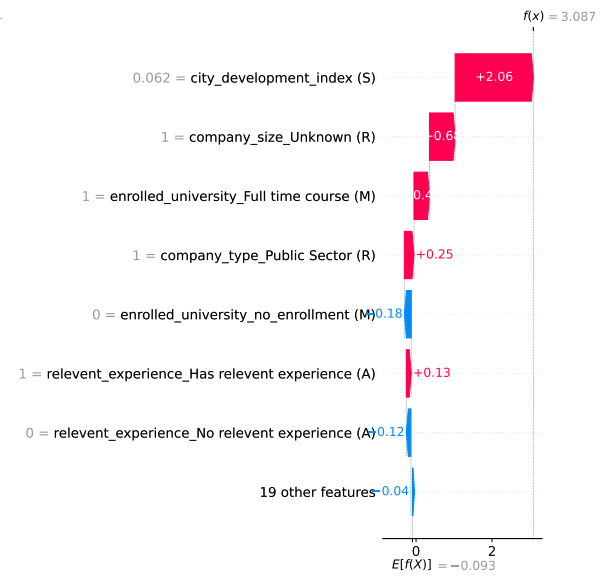
(d) F: GAM - Safest

Figure 15: SHAP Local Interpretability: Profiles with Lowest Attrition Probability (Data Science)

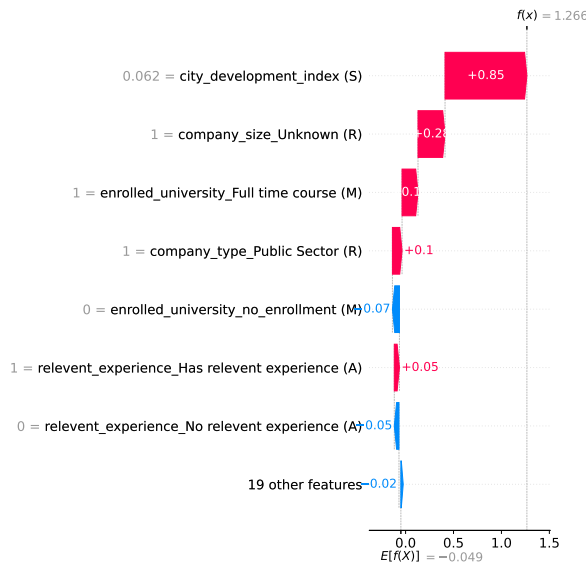
The SHAPs profiles of the most likely leavers rendered in Figure 16 underscores the dangers of *Dual-Track Burnout* as across the **linear models**, **1 = enrolled_university_Full time course (M)** is a prominent **red bar** driving the employee's **risk profile** into the *danger zone* which evinces an incapacity among *data scientists* to meet the demands of a *professional career* and a rigorous academic curriculum if both are to take place concurrently. Therefore, *Enterprise HR* must flag full-time students as imminent *flight risks* as the cognitive load of this highly specialized, computationally heavy profession leaves no bandwidth for a full academic course load.



(a) A: Std Logistic Regression - Riskiest



(b) B: Reg. Logistic Regression - Riskiest



(c) C: Linear SVM (Tuned) - Riskiest



(d) F: GAM - Riskiest

Figure 16: SHAP Local Interpretability: Profiles with Highest Attrition Probability (Data Science)

1.3.6 Model Performance Matrix (Data Science Dataset)

Table 3: Consolidated Algorithm Performance and Hyperparameters (Data Science Dataset)

Rank	model name	tuned hyperparameters	Retention Recall (Class 0)	Attrition Recall (Class 1)	F1- Score (Weighted)
1	D: Decision Tree	{‘min_samples_split’: 7, ‘min_samples_leaf’: 30, ‘max_depth’: 7, ‘criterion’: ‘log_loss’}	0.737	0.788	0.765
2	F: Generalized Additive Model (GAM)	{‘lam’: 0.1}	0.745	0.778	0.768
3	B: Regularized Logistic Regression	{‘solver’: ‘saga’, ‘penalty’: ‘l2’, ‘C’: 555.556}	0.694	0.777	0.733
4	C: Linear SVM (Tuned)	{‘C’: 0.0379269019073225}	0.7	0.77	0.735
5	E: k-Nearest Neighbors	n_neighbors=3	0.84	0.464	0.744
6	A: Standard Logistic Regression	Baseline (Untuned)	0.937	0.264	0.735

Table 3 demonstrates the unsuitability of standard **untuned and unoptimized models** on complex and imbalanced datasets via **Model A**’s dismal **Attrition recall** of **0.264** which implies the **model** fails in identifying 3 out of 4 employees at risk of *attrition* which affirms heavily the **ensemble approach** undertaken by this study.

A similar deficiency was recorded in the performance of **Model E (kNN algorithm)** which despite offering fascinating conceptual insights into *hyper-localized peer groups* ultimately floundered in **global predictive power** by securing an **Attrition Recall** of only **0.464**. This shows that while concluding that *turnover* may be inferred by a *data scientist’s* three closest peers may be intuitive and digestible to *HR practitioners* it is not robust enough to serve as an *enterprise wide early warnings system* as it pertains to this dataset which is further substantiated by the observed outperformance by hard the bifurcations of the **Decision Tree** and the splines of the **GAM**.

The aforementioned **Decision tree (Model D)** emerges as the victor of the contest between **predictive models** by achieving the highest **Attrition Recall** in the panel at **0.788** while also maintaining a highly respectable **Retention Recall** of **0.737**. This means the **decision tree** flags almost 80 percent of *flight risk* without overcommitting *HR* to specious *retention meetings* and also adds to the validity of the *attrition kill zones* conceptualized in Section 1.3.4.

Model D is not alone in this efficacy and is closely followed by **Model F (Logistic GAM)** in performance and secures the highest F_1 score of the **ensemble** of **0.768** in tow with a **recall** only 1% lower than **Model D**'s at **0.778**. This performance equips *enterprise HR* with a more conservative alternative than **Decision trees** if so desired by offering a near identical computational and predictive safety net.

1.4 Limitations and Confounds of Study

The Pipeline conceptualized and built in this study is subject to a following limitations and confounds:

- **Synthetic Baseline Confound (IBM Dataset):** Because this data set was fashioned by *Data Scientists* and wasn't compiled from documented or recorded human behavior it produces an unnatural mathematical cleanliness that cannot be replicated in *enterprise HR practice* owing to the irrational, chaotic "noise" of *real-world workforce dynamics* which often leads to messy data replete with **missing values** and incorrectly entered data.
- **Cultural Translation Pitfalls(Babushkin Dataset):** Though not synthetic (like the IBM dataset) as it is an extraction from an Russian HRIS system, there yet exists a deep structural sparsity in this dataset, the cultural confound identified here is most saliently observed with variables like **greywage** that document endemic economic conditions not present globally. By coercing these localized variables into universal buckets like "*Situational Factors*", the pipeline risks misinterpreting culturally specific *survival tactics* as standard *corporate behaviors* which could result in misdirection if adopted by multinational or foreign enterprises.
- **Administrative Masking of Information Gaps (Data Science Dataset):** As introduced in Section 1.3, **missing or null valued data** was subjected to a **label imputation strategy** where every instance of **null values** in **categorical variables** was replaced with the "**Unknown**". While this *data conservation practice* enabled the garnering of insights such as **company_size_Unknown** being discovered as a massive *flight risk signal* this **imputation strategy** is subject to the risk of **misclassification**. This is because treating the absence of data as a definitive behavioral feature assumes that the data is missing on purpose and dismisses the rather significant possibility of *organizational or departmental indolence, human error* and *data corruption* being the causes of missing categorical information.

- The Static Illusion of Cross Sectional Data (Global Constraint): As all 3 datasets operationalized by this study provide *cross sectional snapshots* rather than *longitudinal workforce assessments*. This could lead to costly oversimplifications as human behavior and *motivation* are not static. If algorithms are only able to adapt to snapshot data they cannot account for the trajectory of features over time such as *morale*. Using the IBM dataset as an example, if an employee had a **JobSatisfaction** score of 2 (on a scale of 0 to 4), this would suggest to the reader that the employee is a *flight risk* if the variable is interpreted under temporal isolation, however had there been additional context that the same employee’s **JobSatisfaction** from a prior period was a score of 1, their score of 2 in the current period would instead divulge the incremental success of *retention efforts*.

1.5 Conclusion

The defining characteristic of this study was the integration of the **MARS Model** (an *Organizational Behavior Framework*) with **Attrition Classification analysis** and subsequently the preferential integration of interpretable **Classification models** over their **black box** counterparts. The former aim which married the principles of *data science* to those of *data management*, allowed for the diagnosis of complex *human centric phenomena* like *Dual-Track Burnout* (Section 1.3.5), *Infrastructural Traps* (Section 1.3.3), and the dangers of “greywages” (Section 1.4) which would have been imperceptible under the auspices of standard **numerical analysis**. The latter aim served to orient the output of this study in a praxis forward manner to the realities of *enterprise HR*. While **Blackbox models** offer substantially near perfect raw scores (i.e Kaggle **random forests** have achieved F_1 scores as high as 0.97 on the Babushkin dataset), the benefits of having reasonably well performing (Models in this study generated F_1 scores between **0.400 and 0.857**) interpretable **models** confers upon the evaluator additional correlational insights that are crucial. Simply but, **blackboxes** inform us “who” will leave but cannot generally inform us why. Hence the adoption of interpretable **models** enables **algorithmic logic** to better conform to business needs by providing transparent and operational logic rather than opaque prophecies.

Furthermore, **Decision Trees (Model D)** and **Logistic GAM’s (Model F)** emerged from the **modelling ensemble** as the most viable *enterprise tools* of the lot. In their isolation of otherwise subterranean *flight risks* (via **Decision Tree** “*Kill Zones*”) and their mapping of behavioral inflection points (via **GAM PDP** “*Sweet spots*”) they affirm that *HR departments* can proactively target *attrition* without relinquishing their data to unexplainable **algorithms**.

The future scope for the enhancement of this methodology can be undertaken from multiple paths. One path would lie in transcending the use of cross sectional data operationalized throughout this study via the compilation and standardization of longitudinal or time series datasets which secures the ability to track the “*temporal velocity*” of an employee’s *morale* and *performance* over multiple quarters, rather than judging them on a single, static day.

A second more action oriented path would involve the embedding of “*Kill Zone*” filters in a live *corporate environment* to facilitate pilot assessments of these filters in observing if *HR magnagers* utilizing **MARS** insights might materially reverse adverse *turnover predictions* in real time with *interventional strategy*.

While featuring an **ensemble of models** grounded in different **mathematical perspectives**, other interpretable **models** such as **K-Means Clustering** and **Stochastic Gradient Descent classifiers** may be paneled in future studies to ameliorate computational diversity. Similarly, while the use of techniques like the **constant imputation strategy** utilized in this study produces misinterpretive risk (Section 1.3), future iterations of this pipeline may benefit from the inclusion of advanced **imputation techniques** (like **MICE - Multiple Imputation by Chained Equations**). The diagnostic capabilities of this pipeline may also be bolstered with **Individual Conditional Expectation (ICE) plots** which visualize how a **machine learning model’s predictions** change for each observation when a specific feature varies, offers an alternative or a complementary inclusion to **SHAPs plots** for localized and individual inspections of *attrition risk*.